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***Abstract*—Speech emotion recognition (SER) draws on how something is said and on what is said. This review examines how SER models are explained when they use the acoustic signal, a transcript of the same speech, or both. We searched Scopus, IEEE Xplore, Web of Science and the ACM Digital Library, supplemented by the ACL Anthology, the ISCA Archive and backward and forward citation searching, screened 1211 unique records published between 2021 and 2026, assessed 83 full-text reports and included 61 studies after eligibility and quality assessment. Each study was coded on four dimensions: the relationship between explanation and model, the temporal resolution of the explanation, the explained target and the explanation method. Delivery format, the speech information explained (acoustic, or acoustic and lexical) and explanation evaluation were coded as separate attributes, and evaluation of the explanation was kept apart from recognition performance. Post-hoc analysis of a trained model was the most common relationship (40 studies) and SHAP the most frequent method (24 studies). Explanations were more often aggregated over classes, features, frequency bands or layers (35 studies) than localised within an utterance (19). Only 10 studies explained lexical as well as acoustic evidence, four of them through generated rationales, and none tested whether a decision rested on the words, on the way they were spoken, or on both. Sixty studies gave recognition results, but only 11 tested the explanation itself: 7 measured decision fidelity and 4 involved human evaluation. These patterns held after borderline studies were removed. Future work should test whether explanations reflect the model's decision, express them in units that listeners can perceive, explain what is said and how it is said for the same decision, and evaluate usefulness with users, including for personalised and audible explanations.**

***Index Terms*—Speech emotion recognition, paralinguistics, spoken language, explainable artificial intelligence, interpretability, systematic review, evaluation of explanations.**

I. Introduction

Speech emotion recognition (SER) infers emotional states from spoken language [1]. Spoken emotion is conveyed both by linguistic content and by paralinguistic cues such as pitch, intensity, voice quality and spectral properties [2], [3], and current SER models learn such acoustic patterns directly from representations such as mel-spectrograms [4]. The two channels are complementary: linguistic content tends to carry more information about valence and acoustic variation more about arousal, although the division is not strict. SER has been studied as a component of real-time human–computer interaction [5], and multimodal systems combine the speech signal with transcriptions produced by automatic speech recognition [6].

This review takes SER in this broad sense. It covers explanations of models that recognise emotion from the acoustic signal, from a transcript of the same speech, produced manually or by automatic speech recognition, or from both, including audio-language models that state the reasons for their decision. Studies on written text alone, such as sentiment analysis of social-media posts, are not included, because their text was never spoken and carries no evidence of delivery. The distinction matters for explanation. When a speech–transcript model gives an emotion word a high weight for the corresponding class, the explanation concerns what was said; when it highlights a rising pitch contour, it concerns how it was said. Both are part of the evidence for an SER decision, and a review that coded only one of them would miss how the two are, or are not, related in current explanations.

SER has moved from handcrafted descriptors, such as mel-frequency cepstral coefficients (MFCCs), fundamental frequency, energy, zero-crossing rate, jitter and shimmer, to spectrogram-based convolutional and recurrent networks, attention, self-supervised speech encoders, fusion with transcripts and, most recently, audio-language models that answer in text [1], [4], [6]; multimodal emotion recognition further combines speech with visual, physiological or contextual signals [2], [7], [8]. As these models have grown more complex, they have also become harder to inspect, which is the general motivation for explainable artificial intelligence (XAI) [9]. The XAI literature has also made clear that illustrative examples are not enough to judge an explanation: its quality has to be assessed quantitatively [10], and its effect on understanding, trust and joint human–AI performance has to be studied with users [11].

In SER, Shapley-value attribution has been used to rank acoustic descriptors and to guide feature selection [12], [13], class activation maps and occlusion to locate evidence in spectrograms [14], attention weights to find emotionally salient frames [15], layer-wise probing to trace how an adapted speech model forms its decision [16], occlusion of spoken word sequences to show which phrases move a valence estimate [17], concept activation vectors to test sensitivity to pitch and to the polarity of the transcript [18], and audio-language models to generate rationales that quote the words and name

vocal cues [19], [20]. Fig. 1 places these explanations in the recognition pipeline.

This variety leads to four recurring confusions. First, the representation a model consumes is not necessarily the one its explanation refers to: a model may take the raw waveform while the explanation interprets learned filters. Second, a temporally structured input does not make an explanation temporally resolved. Third, inspecting an internal component, such as attention used for pooling, is different from building an explanation into the model [21], [22]. Fourth, a model that reads a transcript does not necessarily explain the words, and a rationale written in words does not necessarily concern them; it may describe pitch and loudness. Explanations also reach their users in different forms. Most are visual, a growing number are verbal because of large language models, and a few are audible, for instance when salient regions are played back or contrastive examples are synthesised. Audible explanations suit speech particularly well, and sonification offers a further way of communicating explanatory information [23]. Which format works depends on the listener, which raises the questions of personalisation and usability testing [11], [24].

This paper reviews XAI for SER with a coding scheme that keeps these aspects apart. Its contributions are as follows:

- a systematic review of 61 studies, reported in 62 publications between 2021 and 2026, identified from four bibliographic databases, two subject archives and backward and forward citation searching, with eligibility and quality decisions documented for all 83 reports assessed in full text;
- a taxonomy with four independently coded dimensions (explanation–model relationship, temporal resolution, explained target and explanation method), complemented by delivery format, the speech information explained and explanation evaluation as separate attributes;
- a synthesis that separates evaluation of the explanation from evaluation of recognition performance, with sensitivity analyses for borderline studies and for the speech information explained; and
- a research agenda covering decision fidelity, perceptually meaningful units, joint explanation of lexical and acoustic evidence, audible and personalised explanations, and user studies.

II. Related Reviews

Earlier reviews of SER took the recognition pipeline, rather than the explanation, as the unit of synthesis (Table I). Madanian et al. [3] reviewed machine-learning SER with emphasis on data processing, feature extraction and selection, classifiers, speaker-independent recognition and evaluation. Rathi and Tripathy [25] analysed how corpora and features affect recognition performance, and Alhussein et al. [26] combined a systematic review with a meta-analysis of SER in conversations. These reviews discuss features in depth, but not as explanations of trained models.

Johnson et al. [27] conducted a scoping review of XAI for audio and visual affective computing and coded explanation stage, scope, type, objective and evaluation. Their scheme is the closest to ours, but because it spans audio and visual tasks, the temporal resolution of speech evidence and the explained speech representation are not coded as separate dimensions. Jayasinghe et al. [28] reviewed the interpretability of speech emotion features, covering intrinsic and post-hoc approaches such as attention, feature importance, SHAP, LIME, saliency and representation analysis, and organised their synthesis around the features themselves. The present review codes, for each study, what is explained, how, at which temporal resolution, in which format and on which speech information, and it codes the evaluation of the explanation separately from recognition performance.

III. Review Methodology

The review follows the PRISMA 2020 statement [29]. The screening register, eligibility decisions, quality scores, page locators, extraction codes, source counts and the citation-search log are combined in one data workbook, provided as supplementary material, from which every count, table and figure in this paper was computed. A separate quality-assessment workbook gives the criterion-level justification of each score, and the accompanying figure code documents the mapping of corpora to languages and of venues to publication types.

A. Research Questions

RQ1 How are explanations positioned relative to the trained SER model (D1)?

RQ2 What do explanations refer to, acoustic evidence, lexical evidence or both, and at which temporal resolution (D2, D3, speech information)?

RQ3 Which explanation methods are used, and in which formats are explanations delivered (D4, delivery format)?

RQ4 How are explanations evaluated, separately from recognition performance, and in which language and elicitation contexts?

B. Information Sources and Search

The core sources were Scopus, IEEE Xplore, Web of Science and the ACM Digital Library. The ACL Anthology and the ISCA Archive were added as subject archives because speech and language venues are not fully indexed by the core databases. Each query combined an SER block (for example, “speech emotion recognition”, “vocal emotion”, “emotion in speech”, and multimodal emotion recognition restricted to speech, voice or acoustic terms) with an explanation block containing general explainability terms and method names such as SHAP, LIME, Integrated Gradients, Grad-CAM, layer-wise relevance propagation, occlusion, counterfactual, rationale and sonification. Scopus was searched in titles, abstracts and keywords, Web of Science in the topic field and the ACM Digital Library in titles, abstracts and keywords; the full strings are given in Appendix A. The searches were last run on 2 November 2026. The eligibility scope was extended after the searches had been run, from acoustic evidence alone to speech

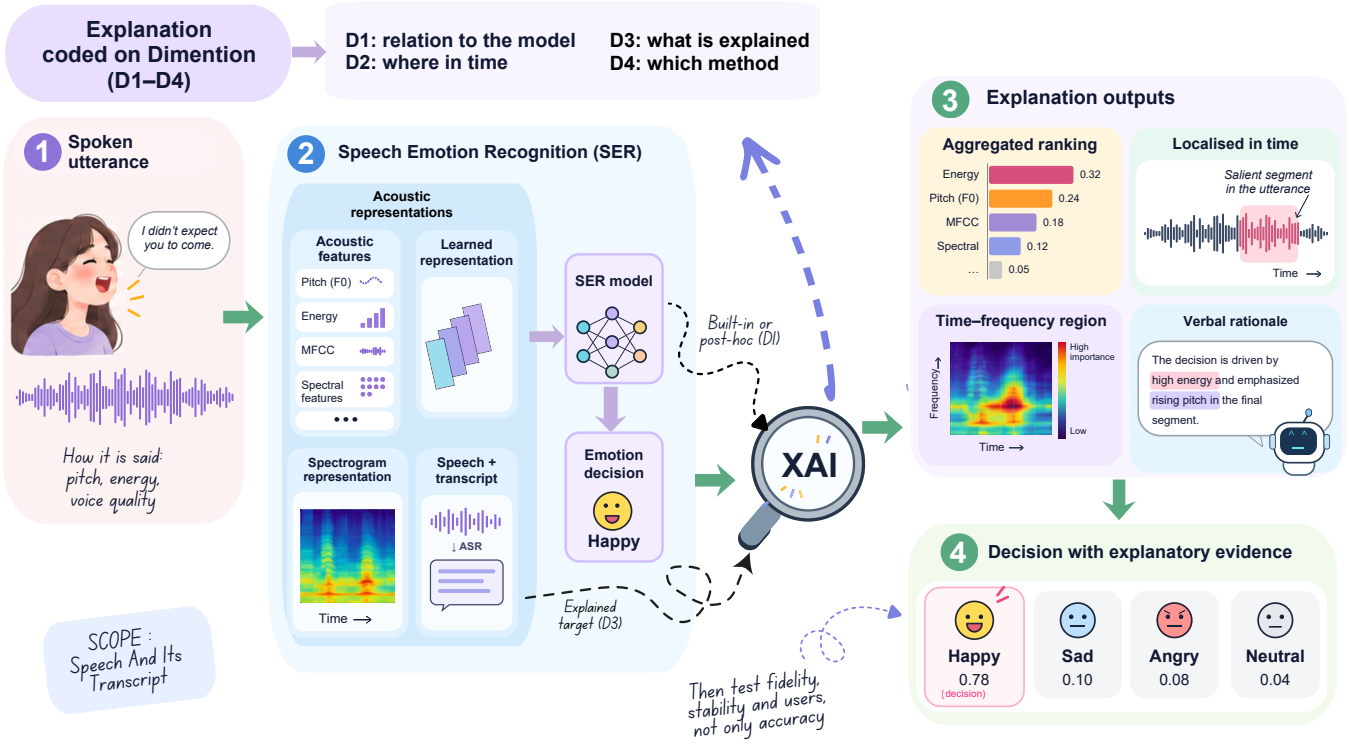


Fig. 1. Conceptual overview of explainable SER as covered by this review. (1) A spoken utterance carries emotion both in how it is said and in what is said; the acoustic signal, a transcript of the same speech, or both are in scope, whereas written-only text and sung material are not. (2) The SER model may be an acoustic model, a speech–transcript model or an audio-language model, and produces an emotion decision. (3) The explanation is coded on four dimensions (D1–D4); it may take the form of a feature ranking, a time segment, a time–frequency region, word weights, a verbal rationale or an audible example, and it may refer to acoustic evidence alone or to acoustic and lexical evidence. (4) Recognition and the explanation are evaluated separately. The utterance and decision are illustrative.

TABLE I
SCOPE OF RELATED REVIEWS RELATIVE TO THE PRESENT REVIEW. ENTRIES SUMMARISE WHAT EACH REVIEW ANALYSES; THEY DO NOT IMPLY THAT A TOPIC IS ABSENT FROM THE REVIEW.

Review	Type and scope	Primary unit of analysis	Treatment of explanation
Madanian et al. [3]	Systematic review; machine-learning SER	Data processing, feature extraction and selection, classifiers, speaker-independent recognition and evaluation strategies	Explanation is not a dedicated analytical dimension.
Rathi and Tripathy [25]	Review; SER corpora and features	Influence of corpora and speech features on recognition performance	Feature relevance is discussed with respect to performance, not as model explanation.
Alhussein et al. [26]	Systematic review and meta-analysis; conversational SER	Annotations, modalities, features, databases and recognition performance	Explanation is not coded.
Johnson et al. [27]	Scoping review of XAI in audio and visual affective computing	Explanation stage, scope, type (feature, example, concept, rule, modality), application objective and evaluation	Structured coding, but across audio and visual tasks; the temporal resolution and the explained speech representation are not separate dimensions.
Jayasinghe et al. [28]	Review of interpretability of speech emotion features	Speech features, their extraction and their interpretation; intrinsic and post-hoc interpretability	Discusses attention, feature importance, SHAP, LIME, saliency and representation analysis, organised around features.
This review	Systematic review; SER from speech audio and its transcript	Report-level extraction and quality assessment, study-level synthesis of what is explained, how, at which temporal resolution, in which format, on which speech information, and how the explanation is evaluated	Four coded dimensions; delivery format, acoustic or lexical evidence, and evaluation coded as separate attributes.

and its transcript. The strings were not changed, because the SER block already contained speech-restricted multimodal terms, and all 597 screened records were re-screened against the extended scope. Studies indexed only under terms for emotion recognition in conversation or multimodal sentiment may nevertheless be under-represented.

Records were matched across sources by normalised DOI or, where no DOI was available, by normalised title. Because database-specific filters were not applied consistently across sources, the publication window and the document-type criterion were applied uniformly after matching. The 2021–2026 window was chosen to capture current practice, not to mark the beginning of XAI research on speech emotion. Coverage of 2026 is necessarily partial.

Backward and forward citation searching was run in OpenAlex [30] on 7 November 2026. The seeds were the 57 reports of the studies included from the database search, the two closest reviews [27], [28] and two closely related preprints [31], [32]; one included report was not indexed in OpenAlex. Each query retrieved the references (backward) or the citing works (forward) of up to three seeds and was restricted to records with "emotion" in the title or abstract. Records published before 2021, duplicate links, records already identified by the database search and preprints were removed before screening, and the remaining records were screened, assessed and appraised with the same criteria and quality rubric as the database records. All queries, responses and decisions are listed in the data workbook.

C. Eligibility

Eligibility was decided at full text against the criteria in Table II. Peer-reviewed journal articles and full proceedings papers were eligible; preprints were not. Where a work existed in several versions, the peer-reviewed version was used, and a conference paper with a later journal extension was assessed as a separate report but counted with the extension as one study. Preprints could not enter the synthesis. Where a preprint bears directly on a point in the Discussion, it is cited there as supporting literature and is not counted in any result. Papers on general audio or signal explanation were eligible when their emotion-recognition experiment and its explanation could be separated, and only that part was extracted. Speech–transcript models were eligible when the text was a transcript of the analysed speech, whether the explanation concerned acoustic evidence, lexical evidence or both. Audio-language models were eligible when they produced an explanatory output with the emotion decision. Systems that also used video or physiological signals were eligible only when the speech or transcript part of the explanation was reported separately; models of written text, including speech synthesised from written text, were not.

TABLE II
ELIGIBILITY CRITERIA APPLIED AT FULL-TEXT ASSESSMENT

Code	Criterion
Inclusion (all required)	
E1	Journal article or full proceedings paper published 2021–2026; preprints, abstracts and theses are not eligible.
E2	The task is recognition of emotion (categorical or dimensional) in spoken speech.
E3	A specified SER model, or a subsystem that contributes to its decision, is explained.
E4	The explained evidence comes from the speech: its acoustic signal, a transcript of the same speech, or both. In models that also use visual or physiological input, the speech or transcript part of the explanation must be reported separately.
E5	At least one empirical explanatory output (global or local) is reported.
Exclusion (one primary reason per report)	
X1	Feature selection or class separation only; no feature, cue or component is linked to an emotion decision.
X2	An explanation module is described, but no eligible explanatory output is shown.
X3	Sung material only.
X4	The speech or transcript evidence cannot be separated from fusion with visual or physiological input.
X5	The explained target is training difficulty rather than the emotion decision.
X6	The text is not derived from spoken speech (written text, or speech synthesised from written text).

D. Selection

Fig. 2 summarises the selection. Of 1301 source occurrences, 505 were duplicates and 199 fell outside the publication window, leaving 597 unique records for title and abstract screening. One of these records, the CHI 2022 study by Zhang and Lim [33], had been excluded at title and abstract screening; it was reassessed in full text and included. Of the 81 reports sought, 4 could not be retrieved, and 77 were assessed in full text. Eighteen were excluded at eligibility, each with one primary reason (Appendix C); the most common reason (8 reports) was that the explanation of a speech, text and video system could not be attributed to the speech or its transcript. The remaining 59 reports were appraised for quality, 2 fell below the threshold, and 56 studies were included from this route. Citation searching yielded 1450 citation links (1092 backward, 358 forward). After removing 551 links to works published before 2021, 197 duplicate links, 75 records already identified and 13 preprints or non-reports, 614 records were screened and 11 reports were sought. Five of these could not be retrieved; the other six met the eligibility criteria, one of them fell below the quality threshold, and five were included [34]–[38]. A conference paper and its journal extension by the same group report the same feature-boosting study; the conference version was treated as a companion report and the study was coded from the journal extension [12], [39], so the synthesis covers 61 studies reported in 62 publications. Two included studies, both on audio-language models [20], [40], were found only in the ACL Anthology. Full-text decisions were not made independently in duplicate, a limitation we return to in Section VI.

E. Quality Assessment

Reports that met the eligibility criteria were appraised with seven criteria (QA v2, Table III), each scored 0, 0.5 or 1. A report passed when its total was at least 4 of 7 and Q1 (task and data origin), Q4 (explanation approach) and Q5 (explanation evidence) were all above zero, because these three criteria define the object of the review; reports that did not pass were excluded. The threshold of 4 requires more than half of the maximum score in addition to the three core criteria: a report can pass with incomplete reporting of its data, protocol or recognition results, but not without an identified emotion task, an explanation method and explanation evidence. Because any cut-off is a convention, the threshold was kept as defined and the main counts were repeated with a stricter threshold of 5 (Table VI). Of the 65 appraised reports, 62 passed and 3 failed. The 62 reports that passed include both reports of the feature-boosting study, which enter the synthesis as one study. Of the three that failed, a LIME explanation over anonymous

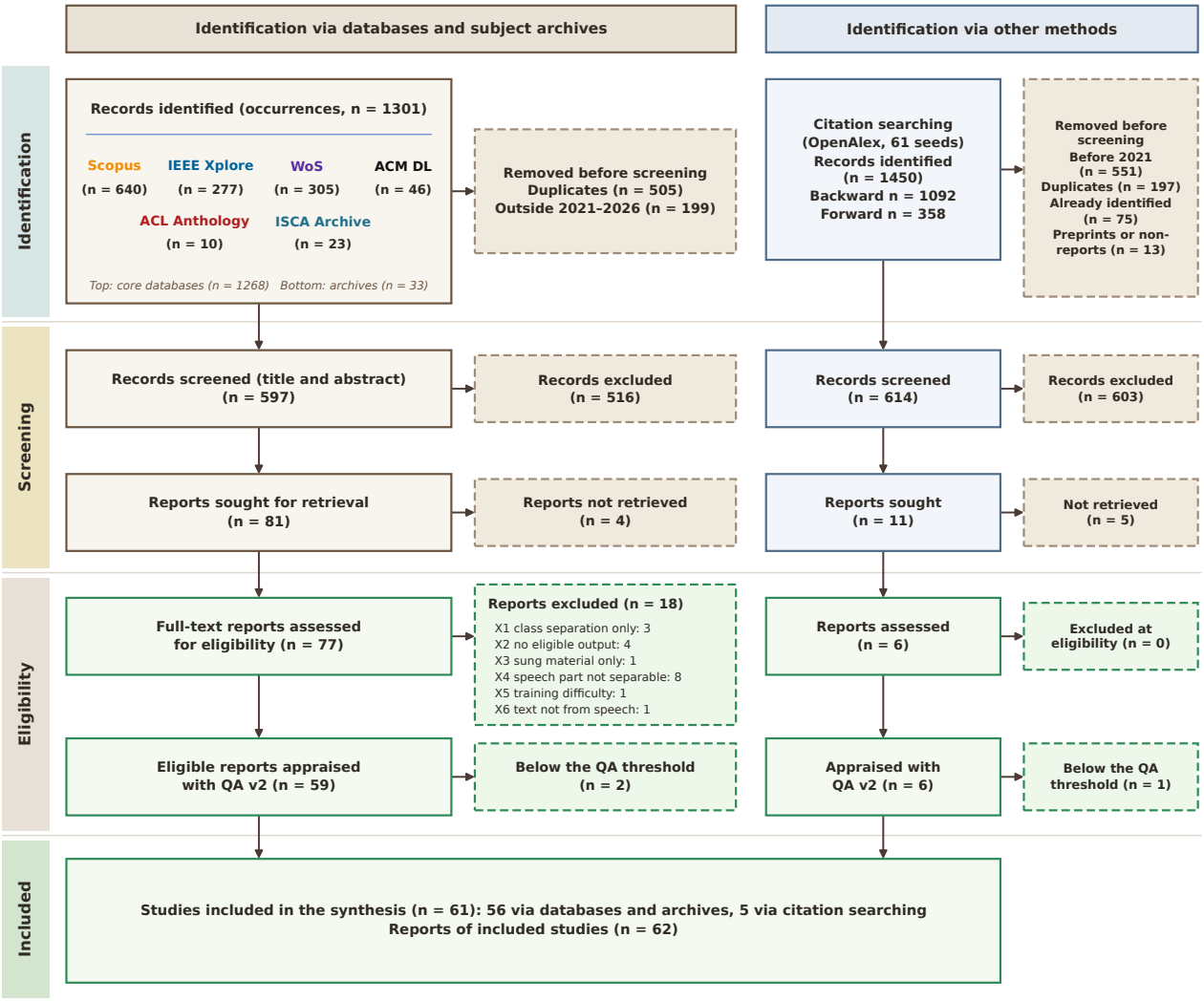


Fig. 2. PRISMA 2020 flow diagram of study selection. Left: database and subject-archive searches, where counts refer to source occurrences until matching, to records during screening and to reports thereafter. Right: backward and forward citation searching in OpenAlex, where counts refer to citation links until matching. Quality assessment (QA v2) was applied after eligibility in both branches.

input indices with no stated test procedure [41] and a retrieval-based rationale with class-level cue phrases [42] both scored 3, and a LIME analysis of an audio-visual system that specified neither the explained model nor the evaluation protocol scored 3.5 [43]. The included studies had a mean total of 6.0; five scored below 5 [37], [44]–[47], among them one that passed without a stated evaluation protocol (Q3 = 0) [44]. Apart from Q7, reporting was weakest for the evaluation protocol (Q3), where partitions were unstated, contradictory or leaking, for example through augmentation or segmentation before the train–test split. Q7 was the weakest criterion overall: only 11 included studies met it in full. All individual scores are listed in Appendix C.

The total describes reporting quality, not the validity of an explanation. Q7 distinguishes interpretation from a test of the explanation, but adding it to task and reporting criteria does not turn the sum into a validity measure. We therefore code explanation evaluation as a separate attribute (Section IV), and none of our conclusions rests on a high quality score. Three questions are kept apart throughout: whether a report belongs to the defined task, whether its methods and outputs are adequately reported, and whether its explanation has been tested. A paper can describe its data and architecture clearly yet show no eligible explanation, which is why eligibility was decided before quality; conversely, a report with a modest total can still contribute a well-defined explanation, as long as it passes the threshold.

F. Data Extraction and Coding

For each included report we recorded the explained model, the corpora in the evaluated pipeline, the explanation methods, the explained target, the observed output with its page, figure or table, and any test of the explanation. Codes were assigned from the explanation actually reported, not from claims in the abstract. Coding proceeded from the explanatory object outwards. We first identified which model output was being explained and which artefact the article displayed: a ranking of named features, an index-based weight vector, a spectrogram

TABLE III
QUALITY-ASSESSMENT RUBRIC (QA V2), APPLIED TO REPORTS THAT MET THE ELIGIBILITY CRITERIA. A REPORT PASSES WHEN THE TOTAL IS AT LEAST 4 OF 7 AND Q1, Q4 AND Q5 ARE EACH ABOVE ZERO. THE TOTAL SUMMARISES REPORTING QUALITY; IT DOES NOT MEASURE WHETHER AN EXPLANATION IS FAITHFUL.

Criterion	0	0.5	1	Decision rule
Q1 Task and data origin	No emotion task on spoken utterances, or origin unclear	Speech-linked task, but spoken origin or labels partly unclear	Emotion recognition from speech, its transcript, or both	Joint emotion and stress targets score 0.5.
Q2 Data and preparation	Corpus or labels not identified	Corpus named; classes, sampling or preparation partly unclear	Corpus, classes, preparation and sampling reported	Wrong corpus facts score 0.5.
Q3 Evaluation protocol	No training or test procedure	Partition ambiguous, contradictory or leaking (e.g., augmentation before the split)	Partition, evaluation unit, model and test protocol replicable in principle	Speaker overlap is a limitation, not a missing protocol.
Q4 Explanation approach	Only a claim of explainability, or attention used for pooling	Method described, but its link to the explained model is unclear	Concrete method applied to an identified emotion decision or model component	Explanations computed on a surrogate or on a different model score 0.5.
Q5 Explanation evidence	No explanatory output	Narrative, a single illustration, or output with unclear link to the decision	Figure, table, concept score or rationale linked to emotion decisions	Confusion matrices, training curves and architectures are not explanations.
Q6 Recognition assessment	No metrics	Metrics with incomplete protocol, comparator or consistent numbers	Appropriate metrics on stated test data with comparisons	Coded independently of the explanation.
Q7 Interpretation and reliability	None	Interpretation or illustration without a test	Interpretation with a fidelity, stability, robustness, human or rationale-grounding test	Agreement between two explanation methods is not counted.

region, a frame or word sequence, an internal representation, a concept score, a verbal description or an audio example. Only then were the relationship to the model, the temporal resolution and the method assigned. A temporal architecture or a sequential input was not taken as evidence of temporal localisation. Methods were coded at the level of named techniques (multi-label); a dominant method was recorded only for a sensitivity analysis, and a report presenting outputs with different targets was coded as mixed. The speech information was coded as lexical when the displayed explanation referred to words, transcript features or transcript content, including words quoted in a generated rationale; a transcript in the model input alone was not enough. Language and elicitation were established from the documentation of the named corpora, so that a report that does not state its language but uses EmoDB is coded as German. Coding was not duplicated independently, so no inter-rater agreement statistic is reported.

G. Synthesis and Sensitivity Analyses

Extraction, coding and quality assessment were carried out per report, whereas the synthesis counts unique studies after companion reports are merged; tables and figures of results therefore give numbers of studies. Counts are not summed where categories overlap. Each study is coded from one report; the only study with two reports is the feature-boosting study, whose conference version [12] was merged with its journal extension [39]. Benchmark corpora are reused across studies, so the counts describe research practice rather than independent samples. We repeated the main counts (i) for the studies whose explanations concern acoustic evidence only, which makes the results comparable with an acoustic-only review, (ii) after excluding two studies whose RAVDESS subset could not be confirmed as speech-only [33], [48], (iii) after excluding two cross-domain audio studies in which emotion recognition is one application among several [49], [50], (iv) after exclusions (ii) and (iii) together (core set), and (v) with a stricter quality threshold, keeping only studies with a QA v2 total of at least 5.

IV. A TAXONOMY FOR XAI IN SER

The taxonomy (Fig. 3) describes the explanation that a report actually presents. It consists of four dimensions, referred to below as D1 to D4, and three further attributes: delivery format, the speech information explained and explanation evaluation. Each report receives one code for D1, D2 and D3, and one or more codes for D4. It is an operational classification for synthesis, not a theory of fundamentally distinct mechanisms; several method families overlap, so methods are coded individually and grouped into families only for reporting. Table IV shows how the difficult cases were decided.

The taxonomy has a primary and a secondary level. The primary level consists of D1–D4, which characterise the explanation mechanism and are the basis of the main synthesis. The secondary level consists of attributes, grouped by what they describe. Evidential attributes describe the explanation as reported and are coded with D1–D4: delivery format, the speech information explained and explanation evaluation. Contextual attributes describe the experiment in which the explanation was produced: corpus, language, elicitation and family of the explained model. They are reported alongside the taxonomy (Fig. 10) but are not part of it, because they characterise the study rather than the explanation, and they are used to qualify, not to classify, the explanations.

A. Dimension 1 (D1): Explanation–Model Relationship

D1 records where the explanation comes from relative to the trained recognition model [21], [51]. Post-hoc analysis applies an explanation method to a trained model or one of its subsystems. Inspection of a built-in component reports the values of a learned component such as filters, graph weights, attention weights or decision coefficients. An explanation

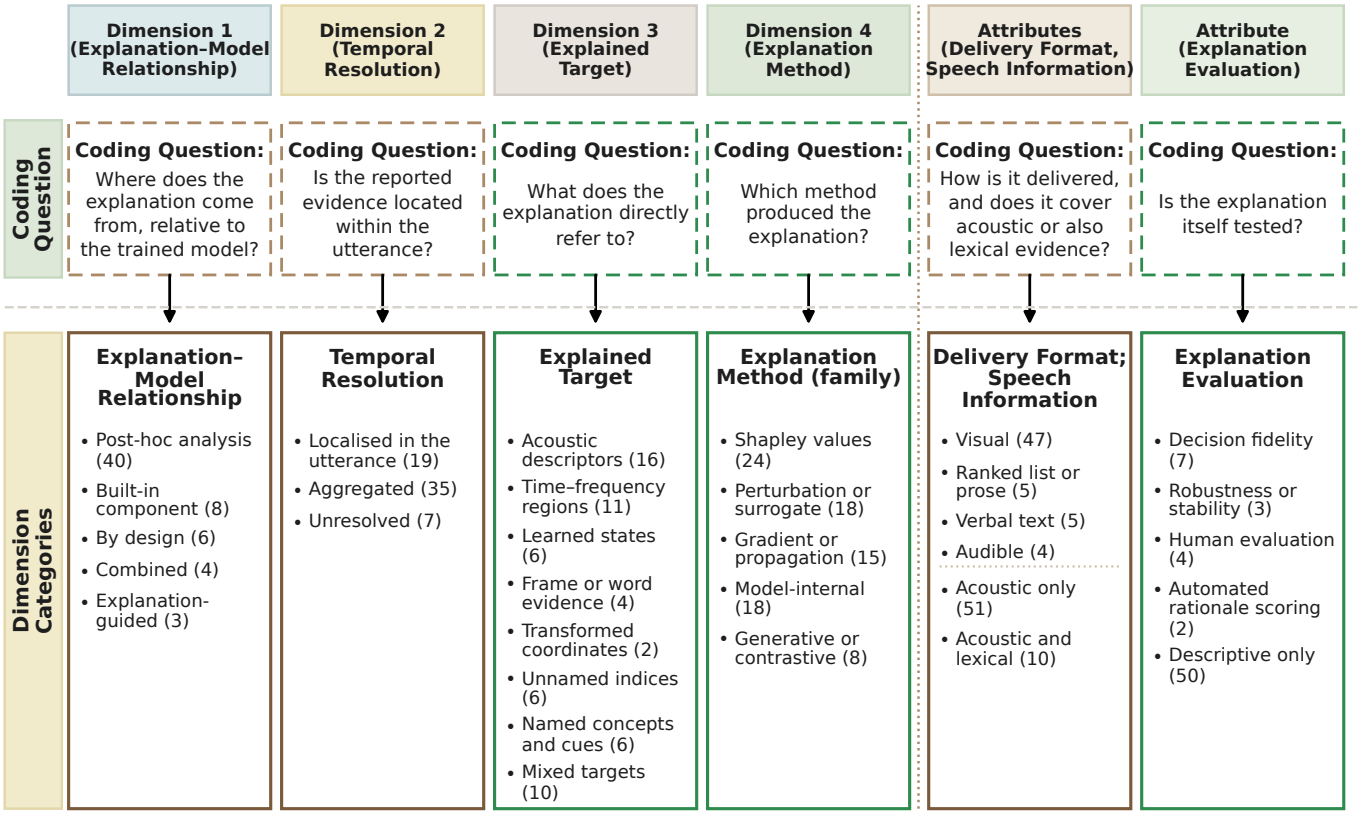


Fig. 3. Taxonomy of explanations in SER. Dimensions D1–D4 are coded independently; delivery format, speech information (acoustic, or acoustic and lexical) and explanation evaluation are separate attributes. Numbers in brackets are included studies; methods are multi-label, so D4 and the evaluation attribute overlap.

generated by design is produced as part of the system output, for example a rationale, a caption or a retrieved example. Two further codes cover the remaining situations: combined reports use a built-in component and a post-hoc method together, whereas explanation-guided reports feed post-hoc explanations back into feature construction or selection.

A component that only serves recognition is not an explanation. Attention used for pooling counts only when its weights are analysed and reported. Likewise, a model with an attention layer whose reported outputs are only embedding plots and class-overlap matrices does not provide an attention explanation; one such study, which showed the separation of classes in a fused speech–text embedding without linking any acoustic cue or word to a decision, was excluded (X1) [52].

B. Dimension 2 (D2): Temporal Resolution

D2 records whether the explanation shows where in the utterance the evidence lies. Speech unfolds over time, and an emotion may be signalled by a short segment, such as a rising final contour, rather than by the utterance as a whole. The dimension is also motivated by work on time-series explanation, which shows that attribution methods need to account for the temporal relations between time points [53]. Three codes are used:

- *Localised*: the output marks evidence on the time axis of the utterance, for example frame-level attention, class activation over spectrogram time, segment occlusion or word-aligned evidence [17], [54]. Regenerated audio that keeps the time course of the original utterance was also coded as localised [50].
- *Aggregated*: the evidence is summarised over a class, an item, a feature, a frequency band or a layer, as in a global SHAP summary or a ranking of utterance-level descriptors.
- *Unresolved*: the output cannot be related to time from the information reported, for example local LIME weights on flattened inputs whose positions are not mapped back to frames.

When a report presents several outputs, D2 is coded as localised if at least one of them is localised. D2 is assigned from the displayed output, not from the model. A sequential input, a recurrent or attention architecture, or the learning of static and dynamic representations does not by itself make an explanation temporally resolved; a study whose only output was a class-separation plot was therefore excluded (X1) [55]. D2 is also distinct from the local–global distinction: a local explanation of one utterance can still be aggregated over time if it weights utterance-level descriptors.

C. Dimension 3 (D3): Explained Target

D3 records what the explanation directly refers to, which may differ from the model input. SincNet models take the raw

waveform, but their explanations interpret the learned band-pass filters, so D3 is a learned acoustic representation [56], [57]. Topological features are explained as persistence-image coordinates rather than waveform time stamps [58], and feature-boosting pipelines explain principal components that are traced back to the original descriptors only afterwards [12], [39]. When flattened inputs or selected features are shown only as indices, the target is coded as unnamed indices, because such outputs cannot support claims about named acoustic cues or words [47], [59]–[62]. Two categories are defined so that they apply to both acoustic and lexical evidence. *Frame or word evidence* covers learned weights over frames or over word-aligned segments [15], [54]. *Named concepts and cues* covers human-defined concepts tested in a model's representation, such as voice pitch or the polarity of the transcript [18], as well as cues and words named in a generated rationale [19], [20]. D3 records the form of the target; whether it concerns acoustic or lexical evidence is recorded by the speech-information attribute described below.

D. Dimension 4 (D4): Explanation Method

D4 lists every named method that produced a reported output. For synthesis the methods are grouped into five families: Shapley-value attribution [63]; perturbation and surrogate attribution, including LIME [64], permutation importance and occlusion; gradient and propagation attribution, including Integrated Gradients [65], DeepLIFT with its reference-based formulation [66], Grad-CAM [67], Grad-CAM++ [68] and layer-wise relevance propagation [69]; model-internal inspection, including attention, learned filters, network weights, tree importance, layer probing and concept activation vectors [70]; and generative, contrastive or example-based explanation. The families overlap: Integrated Gradients, DeepLIFT and Grad-CAM are all attribution methods, and Shapley values can be estimated by perturbation. Generative explanation is further divided by what is generated. Verbal rationales and captions differ from regenerated audio; replacing less influential embedding components with audio from an audio-language model and testing whether the decision is preserved is an audio-based fidelity method, not a verbal rationale [50].

The method name alone says little about the acoustic unit of interpretation. SHAP assigns feature contributions within an additive explanation framework [63], and LIME fits an interpretable approximation in the neighbourhood of an instance [64]. Both return a vector of weights, but in acoustic SER that vector may describe utterance-level descriptors, perturbed frequency bands, reduced components or anonymous input positions. Integrated Gradients accumulates gradients along a path from a reference to the input [65]; DeepLIFT also compares against a reference but uses a different propagation rule [66]; Grad-CAM weights a chosen convolutional representation by class-related gradients [67]. A speech explanation should therefore state the input or layer to which relevance is assigned and how that representation maps to the recording; a heatmap over a transformed representation does not acquire a waveform-time meaning through its colour scale. Model-internal quantities, such as filter responses, attention weights, graph connections and decision coefficients, raise a different question again: a filter plot describes spectral preferences, a probe shows what a hidden state encodes, and only an intervention shows that the output depends on it. These differences are why D2 and D3 are coded independently of D4, and why the evaluation attribute records which stronger claims were actually tested.

E. Delivery Format, Speech Information and Explanation Evaluation

The delivery format records how the explanation is communicated: as a plot, map or table; as a ranked list or prose; as verbal text; or as audible output. It is coded separately from D3, because a textual rationale may explain acoustic evidence without making the explained target linguistic.

The speech information records whether the explanation concerns acoustic evidence only, or lexical as well as acoustic evidence. The code follows the displayed explanation, not the model input: a speech–transcript model whose explanation shows attention over acoustic frames is coded as acoustic, whereas an audio-only model is coded as lexical when its occlusion is aligned with spoken words and interpreted by their meaning [17]. No included report explained lexical evidence alone, so the attribute has two values.

Explanation evaluation records whether the explanation itself was tested. Decision fidelity requires that explanatory evidence is removed, masked, inserted or regenerated and that the effect on the emotion decision is measured against a control. Robustness and stability require a comparison of explanations under controlled input perturbation or across repeated runs. Human evaluation requires participants to rate or use the explanations. Automated scoring of generated rationales, for instance by a language model acting as judge of whether the quoted words occur in the transcript, is recorded as a fourth category, because it checks the rationale against the input without involving people. Recognition accuracy, cross-corpus accuracy, augmentation experiments and architecture ablations are recorded as recognition performance, and agreement between two explanation methods is not counted as stability or fidelity.

V. Results

A. Overview

The 61 included studies appeared between 2021 and 2026, with 19 from 2026 and 19 from 2025 (Fig. 4). Thirty-one were journal articles and 30 full proceedings papers. The explained systems were tree-based or other classical classifiers (16 studies), convolutional, recurrent or ensemble networks (18), pretrained encoders and Transformers (10), specialised or mixed architectures (12) and language or audio-language models (5) (Fig. 10d). Seven explained systems received a transcript together with the audio [15], [18], [36], [54], [62], [71], [74]. The publication profile describes the retrieved literature rather than a growth rate, since search coverage, partial indexing of 2026 and the peer-reviewed-only criterion all affect which recent systems enter the corpus. The model category refers to the system that was actually explained: a

TABLE IV
OPERATIONAL CODING DECISIONS ILLUSTRATED WITH REPORT-LEVEL EVIDENCE. FIG. 3 DEFINES THE CATEGORIES; THIS TABLE SHOWS HOW DIFFICULT CASES WERE DECIDED.

Dimension	Potential confusion	Example	Coding consequence
D1 Relationship	An attention or fusion architecture automatically provides an explanation.	Embedding plots and intersection matrices of a speech–text model; no cue or word is linked to a decision [52].	Excluded (X1) rather than coded as an attention-based explanation.
D2 Temporal resolution	A sequential input or a local explanation establishes temporal localisation.	SHAP and LIME weights on unnamed, flattened inputs [47].	A local output exists, but its relation to time is unresolved.
D3 Explained target	The explained target equals the input representation.	Waveform models whose explanations interpret learned filters [56], [57]; SHAP computed on principal components of a fused representation [71].	Coded as a learned representation or as transformed coordinates, not as the waveform or the words.
D4 Method	Every generative model produces a verbal rationale.	Less influential components are replaced with generated audio and the reconstructed decision is tested [50].	Generative audio is kept distinct from verbal rationale generation.
Delivery format	A textual explanation necessarily targets linguistic evidence.	Generated captions that name pitch, volume and rate [72].	Verbal format, acoustic speech information.
Speech information	A transcript in the model input means that the explanation covers lexical evidence.	Audio and transcript encoders, with attention analysed on acoustic frames only [15].	Coded as acoustic; lexical is coded only when words or transcript content appear in the explanation.
Evaluation	Recognition gains or agreement between explainers establish fidelity.	Explanation-guided feature construction [12], [39]; comparison of attribution outputs [73].	Selection gains and inter-method agreement are not counted as explanation validation.

paper may compare several classifiers but explain only one of them.

B. RQ1: Explanation–Model Relationship

Most studies (40 of 61, 66%) applied a post-hoc method to a trained model (Fig. 5). Post-hoc analysis was the most common relationship in every year from 2023 onwards, and explanations generated by design appeared only in 2025 and 2026. Built-in components were explained mainly through model-internal inspection, whereas Shapley, perturbation and gradient methods were applied mostly post hoc. Eight studies inspected a built-in component, such as learned SincNet filters [56], [57], graph adjacency between acoustic parameters [75], attention over frames or features [15], [38], [54], [76] or the class-wise decision coefficients of a representation constrained towards acoustic descriptors [77]. Six generated explanations by design: a speech language model that produces acoustic and content rationales [19], captions with pitch and volume descriptors [72], a distilled chain of thought that reasons over the transcript and verbalised pitch and intensity trends [74], audio-language models trained to state the evidence before the label [20], [40] and a cue-specific saliency network with contrastive audio [78]. Four combined built-in and post-hoc explanations [33], [61], [79], [80], and three used post-hoc explanations to guide feature construction or selection [12], [13], [39], [62].

The labels matter less than what they imply. In the SincNet study, the learned filters belong to the recognition architecture, but their relevance is established after training by masking individual filters [56]. In IG-CNN, acoustic constraints make part of the representation easier to interpret while an unconstrained part remains outside the explanation [77]. Neither case makes the whole decision process transparent. An explanation produced by design may likewise still need an external fidelity test, since producing a rationale together with a label shows that the system can produce rationales, not that they are correct. In the explanation-guided reports, the same Shapley values select features and explain the resulting model [12], [13], [39], [62]. Better recognition after keeping the top-ranked features supports the selection procedure, but it does not show that the original ranking was faithful.

C. RQ2: Explained Target and Temporal Resolution

Acoustic descriptors were the most frequent explained target (16 studies), followed by time–frequency regions or spectral bands (11), mixed targets (10), learned representations or model states (6), named concepts and cues (6), unnamed indices (6), frame or word evidence (4) and transformed coordinates (2) (Fig. 6).

Explanations were aggregated in 35 studies, localised within an utterance in 19 and unresolved in 7. The two dimensions are closely linked. Every explanation of acoustic descriptors or learned states was aggregated, because descriptor-level SHAP summaries, tree importances and filter responses are reported per class or per feature. Localised explanations came mainly from class activation maps, occlusion and attention over spectrograms or frames [14], [15], [17], [33], [36], [38], [49], [50], [54], [61], [73], [74], [78]–[84]. Grad-CAM trajectories, for example, were compared with energy and fundamental-frequency contours [83], and spectral bands were suppressed to record the effect on estimated arousal [17]. The unresolved cases were local explanations of flattened or transformed inputs whose positions could not be traced to time [45]–[48], [59], [85], [86]. Concepts and cues named in generated text were aggregated in all but one report: the rationales describe an utterance as a whole, and only the distilled chain of thought refers to how pitch and intensity change within it [74].

As noted in Section IV, a local explanation is not necessarily localised in time, and a time–frequency map may localise evidence yet remain hard to interpret perceptually. The unresolved category records the missing link between the displayed coordinates and speech time; it does not mean that no attribution was computed. Aggregation is not a weakness in itself either. Class-level descriptor rankings can support corpus

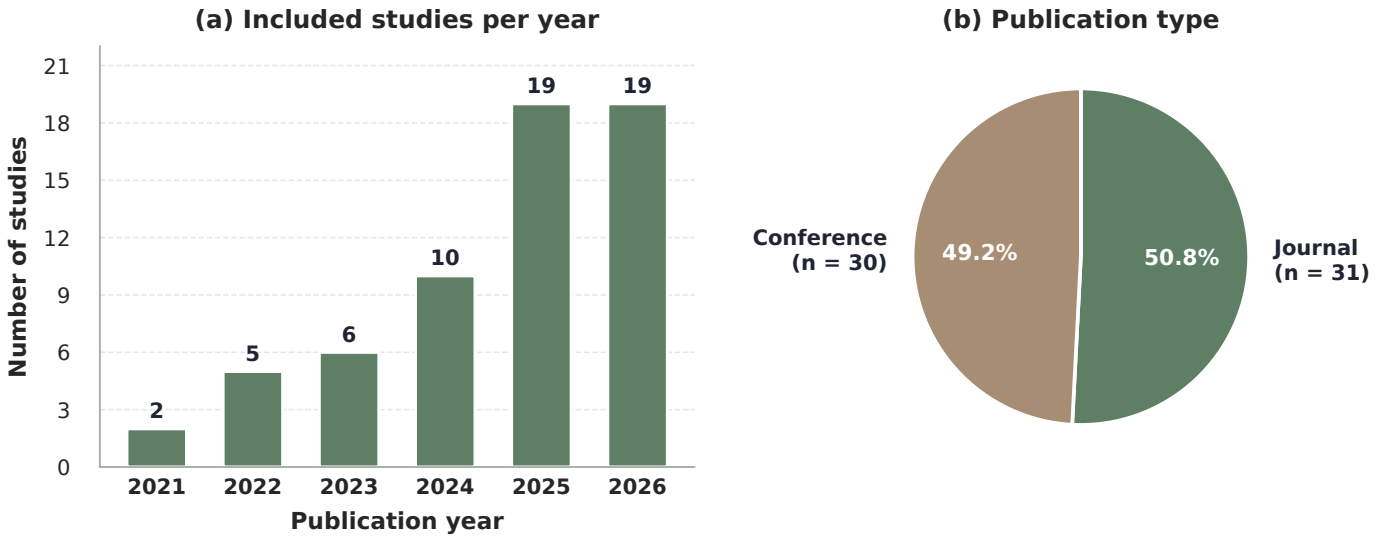


Fig. 4. Included studies by publication year (a) and publication type (b). The 2026 count covers only the months searched; full proceedings papers are counted as conference papers.

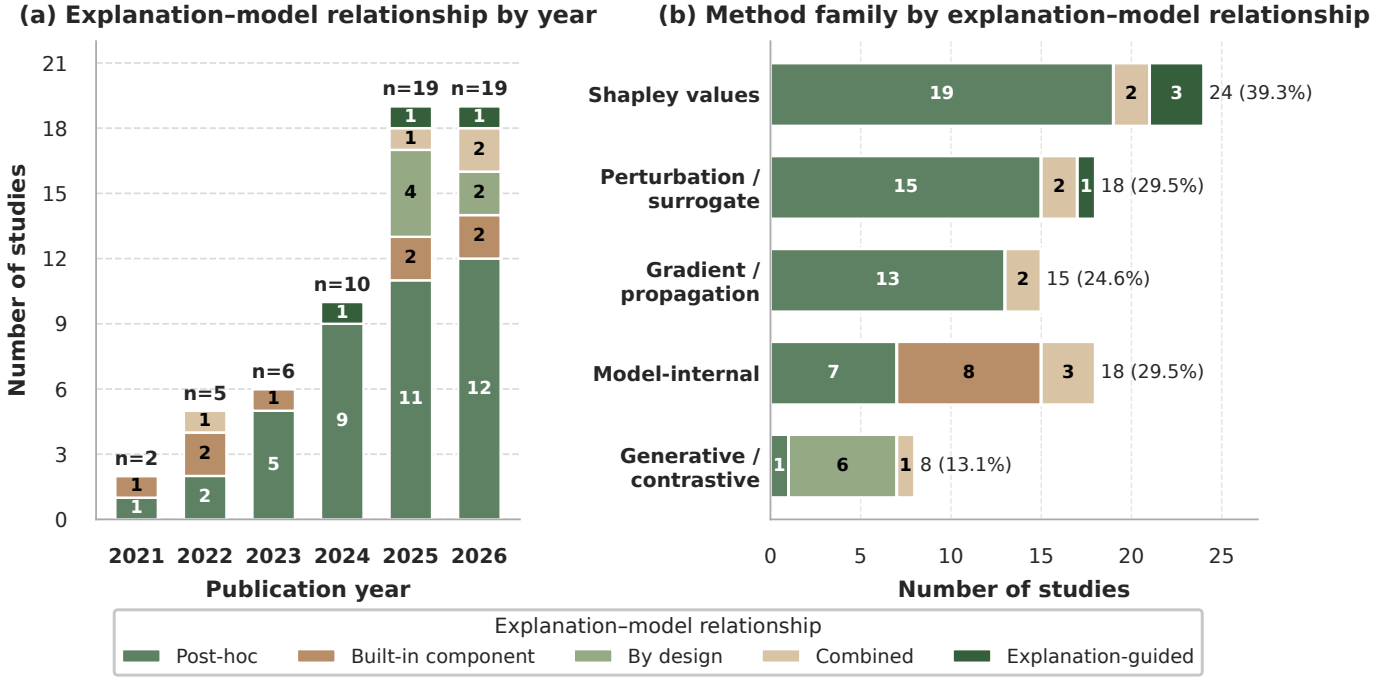


Fig. 5. Explanation–model relationship (D1) by publication year (a) and by method family (b). A study contributes once to each method family it uses, so (b) does not sum to 61.

auditing or model comparison, whereas a listener reviewing one ambiguous decision may need a segment and a contrastive cue. Problems arise when the conclusion goes beyond the representation: an attribution to a principal component is not a pitch attribution, and a band-pass filter response is not a word-level explanation. The feature-boosting and SincNet studies show why the transformation from input to learned representation to explained target has to be reported explicitly [12], [39], [56], [57].

1) Acoustic and lexical evidence: Most explanations concerned acoustic evidence only (51 studies). Lexical evidence appeared in 10 studies [17]–[20], [36], [40], [54], [62], [71], [74], all published from 2022 onwards and 8 of them in 2025 or 2026, and always alongside acoustic evidence. The forms differ widely. Concept activation vectors tested a voice-pitch concept in the audio branch and an utterance-polarity concept in the transcript branch of the same model; polarity was significant for every class in the transcript branch but only for some classes after fusion [18]. GradientSHAP ranked indexed audio and transcript features and reported the share of each modality among the top 20; transcript features made up most of the list for nearly every emotion, with sadness as the exception in one configuration [62]. SHAP over principal components of the fused representation of a speech–transcript

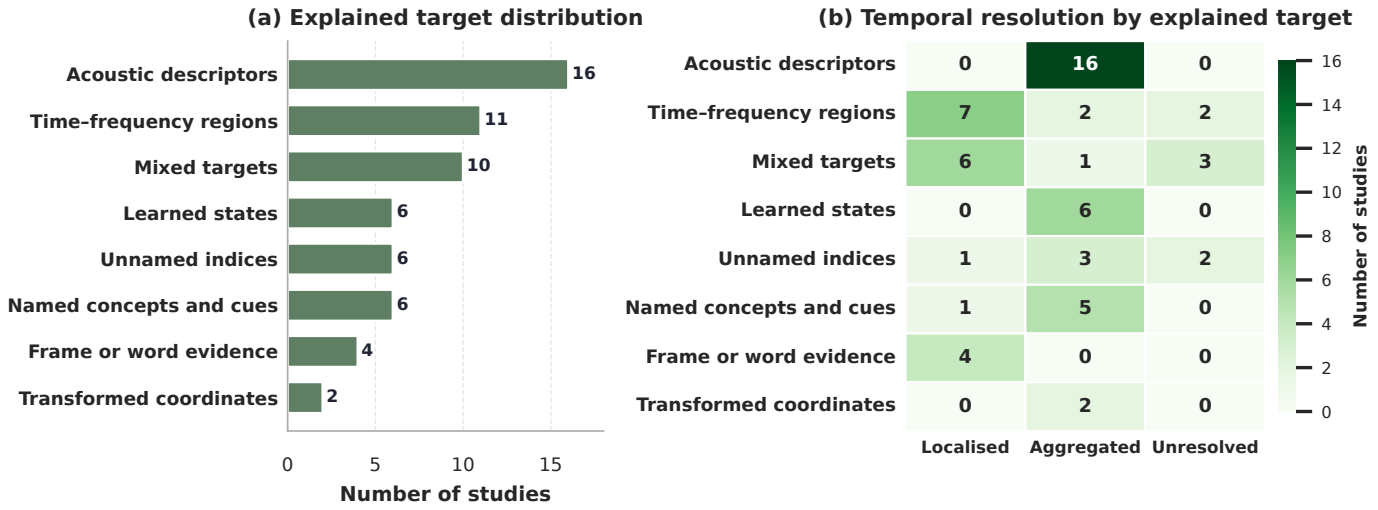


Fig. 6. Explained target (a) and its cross-tabulation with temporal resolution (b). Each study has one D2 and one D3 code.

network showed interactions between MFCC-derived and textual components [71]. Three studies localised words in time: attention over word-aligned segments was compared with human annotation [54], occluding the audio of spoken four-word sequences showed that removing phrases containing "killing" or "terrifying" moved a valence estimate towards neutral [17], and Integrated Gradients was plotted over the waveform and the transcript tokens of a fused RoBERTa–WavLM model, with high token attributions on words such as "great" and "song" in a joy example [36]. The remaining four studies generated rationales that quote or paraphrase what was said and name vocal cues [19], [20], [40], [74].

None of these studies isolated the contribution of the words from that of their delivery for a given decision. Modality shares among the top-ranked features describe a ranking, and they depend on the number of features per modality: in the report above, up to 768 text dimensions competed with 74 acoustic ones [62]. Occluding the audio of a word removes its lexical content and its prosody at the same time [17], a rationale that quotes a word and mentions a raised voice does not show which of the two the model used [20], and parallel token and waveform attributions in a fused model may, as their authors note, reflect correlated content rather than the use of each channel [36]. The lexical studies nevertheless tested their explanations more often than the acoustic-only studies (4 of 10, against 7 of 51), two of them through human or automated assessment of generated rationales.

D. RQ3: Methods and Delivery Formats

SHAP was used in 24 studies (39%), LIME in 10, Grad-CAM in 7, attention analysis in 7 and permutation importance in 5 (Fig. 7). We found 24 distinct methods, 11 of them in a single report. Eighteen studies used more than one method, typically a global method with a local one (SHAP with LIME) or an internal component with a post-hoc method (attention with SHAP or Grad-CAM). Counting only the dominant method of each report did not change the ranking: SHAP

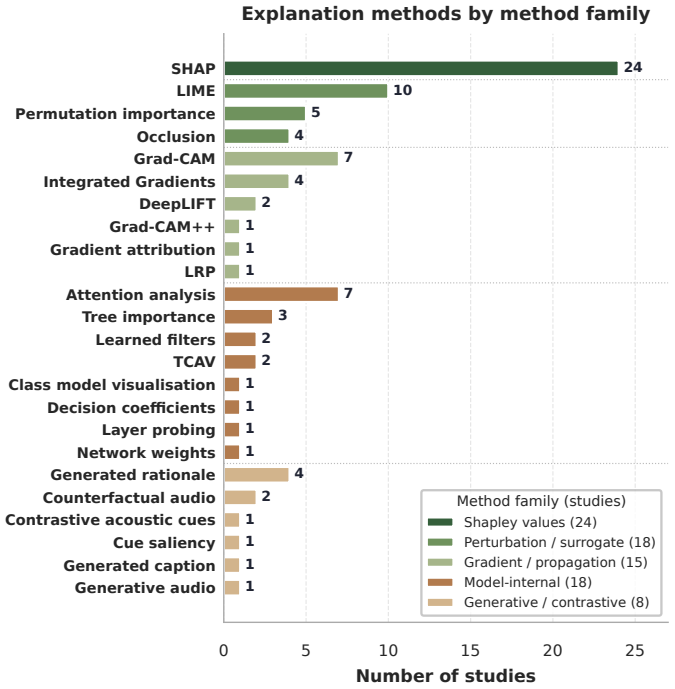


Fig. 7. Explanation methods in the 61 included studies, grouped by method family. Methods are multi-label, so a study is counted once for each method it uses.

remained first (22 studies), followed by LIME (5) and Grad-CAM (5).

The same method can expose very different things depending on the pipeline (Fig. 8). Shapley-value attribution usually ranked engineered descriptors or transformed features, and its acoustic meaning depends on the feature map: SHAP values over named acoustic descriptors [87]–[90] answer a more directly inspectable question than values over principal components [12], [39] or selected indices [47], [60]. The Shapley reports also differ in purpose. The feature-boosting family explains transformed components and traces them back to descriptors [12], [39], and the explainability-guided selec-

tion study combines SHAP with permutation importance [13]; both concern feature construction and model refinement. The analysis of Urdu speech and the gender- and class-specific analyses [88], [89], by contrast, use acoustic attribution to interpret the behaviour of a trained model. LIME likewise spans perturbed frequency bands compared across corpora and models [91], a mixture of embedding and acoustic variables [73] and anonymous flattened inputs [47]; only the first establishes an explicit spectral unit, and the word "local" does not repair the missing correspondence in the last. Shapley values computed on a surrogate add a further step: when SHAP explains a gradient-boosted model fitted to the reduced representation of a speech–transcript network, the explanation describes the surrogate unless its agreement with the network is shown [71].

Gradient and propagation methods usually targeted time–frequency regions or latent representations. Grad-CAM and Grad-CAM++ heatmaps show where a decision is sensitive within a spectrogram [14], [82], and overlays with fundamental-frequency, energy or formant trajectories give such maps a more concrete acoustic reading [83], [84]. Attribution over topological coordinates, in contrast, explains a transformed representation [58]. Scope matters here as well. In one model-comparison report, SHAP explains the tree learners, LIME the voting ensemble and Grad-CAM an encoder [73]; these are explanations of different predictors, and their agreement cannot validate one common decision function. In another, the feature importances come from random forests and decision trees, although neural-model performance is also reported [44].

Model-internal inspection is equally varied: attention weights, filter responses, graph adjacency, layer probes and decision coefficients expose different aspects of a trained system [15], [16], [57], [75], [77], at scales ranging from frequency-selective filters to the internal pathway of a LoRA-adapted Whisper model [16]. A frame-level attention plot may be easy to inspect in time but still needs a reliance test, and a layer probe may clarify internal organisation without identifying a cue a listener would recognise. Concept activation vectors sit between attribution and inspection: they test whether a human-defined direction in a hidden layer, such as high pitch or positive transcript polarity, moves the class score, and the same procedure can be applied to the audio and the transcript branch of one model [18]. Generative methods are separated by what they generate: an audio-language model reconstructs decision-preserving audio [50], whereas speech and audio-language models write a textual account of the emotion [19], [20], [40], and a distilled student reproduces a teacher's chain of thought [74]. All involve generation, but only the first tests whether its output preserves the decision.

Explanations were delivered as plots, maps or tables in 47 studies and as ranked lists or prose in 5. Verbal explanations appeared in 5 studies, all produced by language or audio-language models [19], [20], [40], [72], [74]. Four studies included audible output [14], [33], [50], [78]: class-activation regions converted back to audio [14], regenerated audio that preserves the emotion decision [50], and contrastive or counterfactual speech that lets listeners compare cues [33], [78]. Only the last two tested whether listeners benefited. In the RexNet study, synthetic counterfactuals generated with StarGAN-VC were evaluated only in the model experiments; the user studies used real recordings of the same speaker saying the same sentence with different emotions, selected for correct recognition and coherent explanations [33]. The observed benefit therefore concerns those matched recordings and cue comparisons, not synthetic audio. The controlled experiment recruited 175 participants and excluded 14 who did not play any clip [33].

E. RQ4: Evaluation and Data Contexts

1) Recognition performance versus explanation evaluation: Sixty of the 61 studies gave recognition results, and 41 received the full Q6 score for metrics with a stated protocol (Fig. 9); this score does not certify speaker separation or the absence of leakage. Far fewer tested the explanation. Decision fidelity was measured in 7 studies: by masking individual SincNet filters [56], by insertion curves on a CREMA-D classifier [49], by regenerating less influential components and checking that the decision is preserved [50], by removing high- versus low-attention segments [54], by ablating salient regions in cue-specific and counterfactual explanations [33], [78] and by replacing the frequency profile of an utterance with that of another class and checking that the prediction moved to that class [35]. Robustness of explanations under controlled perturbation was tested in 2 studies [49], [78] and repeated-run stability in 2 [18], [49], one of them by retraining concept vectors and comparing them with random concepts [18]. Human evaluation appeared in 4 studies [19], [33], [72], [78]; two measured whether explanations improved listeners' own emotion judgements [33], [78], and the other two measured acceptance of, or preference for, generated rationales and captions. Automated scoring of generated rationales appeared in 2 studies [19], [20]; in one, a language model judged whether the words quoted in a rationale occurred in the transcript and whether they supported the label, and found most quotes grounded but fewer of them relevant [20]. Overall, 11 studies (18%) contained at least one test of the explanation and 50 (82%) were descriptive only. Tests were rarest for the most frequent methods: only one Shapley study and one perturbation or surrogate study tested decision fidelity, whereas 6 of the 8 studies with generative or contrastive methods tested the explanation in some way (Fig. 9b). Table V sets out what each test can and cannot support.

Two further kinds of evidence were not counted. One report compared attention-derived segments with human annotations [54]; this shows agreement with annotators rather than usefulness, and the report is counted only for its attention-removal fidelity test. Another gave alignment and stability percentages without describing how they were obtained [46]. Agreement between explanation methods was not counted either. SHAP–LIME agreement was close to zero in one report (Spearman ρ between -0.25 and 0.03 over five utterances) [73] and very high in another (Pearson $r = 0.997$ for one anger sample) [61], but neither value shows that the explanation reflects the decision.

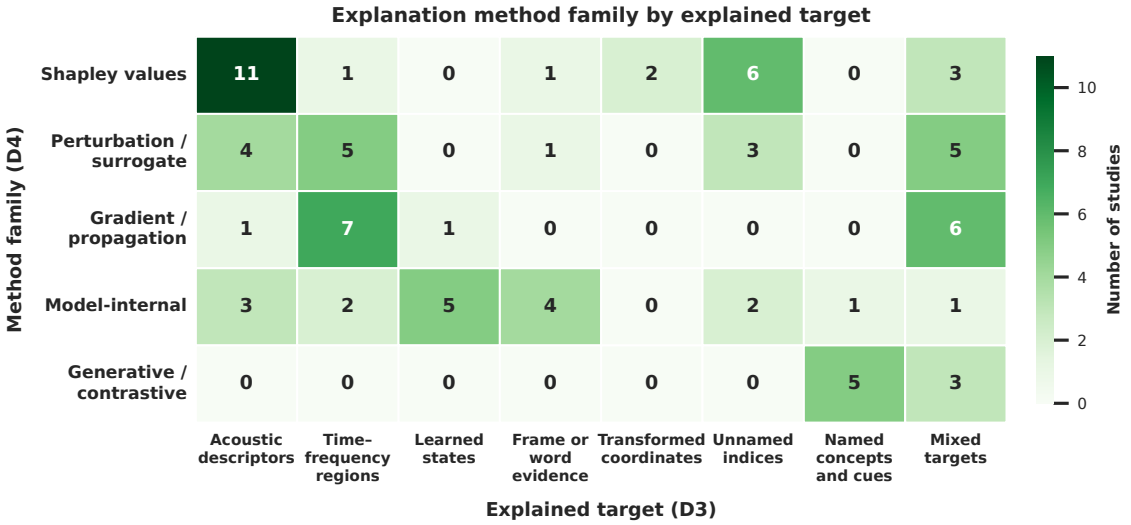


Fig. 8. Explanation method family (D4) by explained target (D3). A study using several method families appears in each corresponding row.

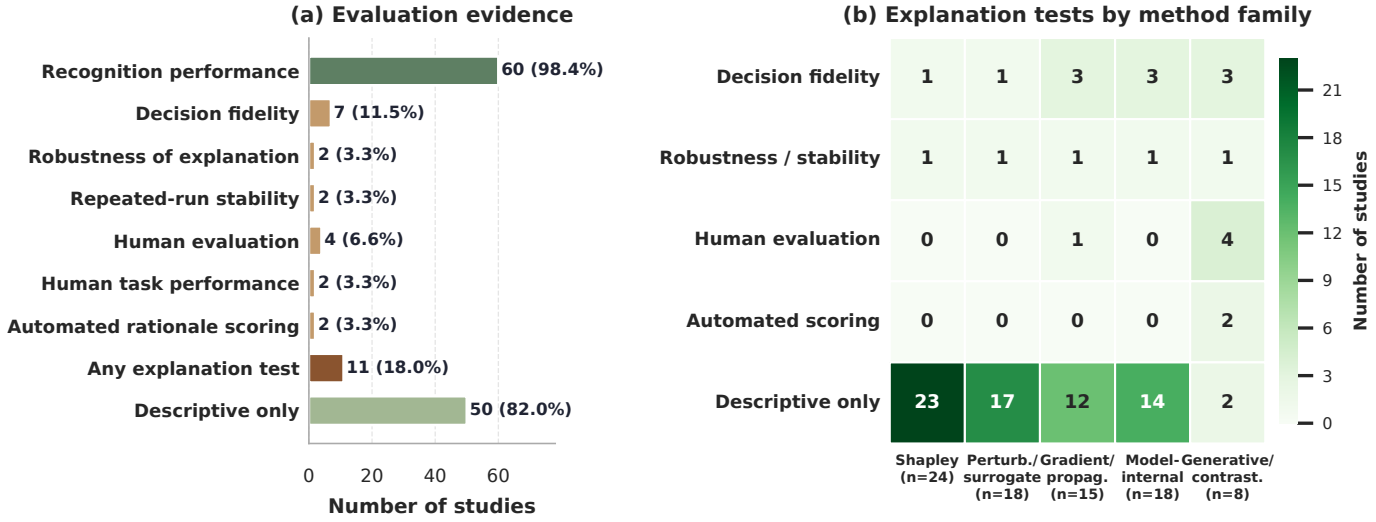


Fig. 9. Recognition performance versus evaluation of the explanation itself (a) and explanation tests by method family (b). Categories overlap; recognition accuracy is not counted as a test of the explanation.

2) Languages and elicitation: Fig. 10 summarises the corpus, language, elicitation and model contexts. RAVDESS [92] was the most frequently used corpus (22 studies), followed by EmoDB [93] (20), IEMOCAP [94] (16), SAVEE [95] (12), TESS [96] (11), CREMA-D [97] (9) and MELD [98] (6). Every study that explained lexical evidence used a corpus distributed with transcripts, most often IEMOCAP or MELD. English corpora appeared in 46 studies and German corpora in 20, all of which used EmoDB, a corpus of acted German sentences [12], [34], [39], [46], [59]–[61], [75], [76], [80], [82]–[86], [89]–[91], [99]–[101]. Across the named corpora, 17 languages were represented. Thirty-four studies used at least one corpus in a language other than English, while 27 used English corpora only. These counts describe the corpora named in each pipeline; for multilingual corpora they do not establish which language subset each experiment actually used.

Acted, scripted speech dominated: 32 studies used only acted, scripted corpora. IEMOCAP, used in 16 studies [16], [18]–[20], [34], [40], [54], [56], [71], [72], [74], [75], [77], [79], [91], [100], contains actors performing both scripted and improvised dyadic interactions [94] and is therefore not conversational speech in the sense of naturally occurring talk. Naturalistic podcast speech from MSP-Podcast [102] appeared in 4 studies [15], [17], [19], [72], television, film and radio material, mostly the MELD dialogues from a television series, in 9, online video monologues in one [62], and spontaneous speech in 4, one of them clinical recordings of French speakers with Huntington's disease [103].

Corpus recurrence cuts both ways. It makes methods easier to compare when label sets, partitions and explained targets are aligned, but it concentrates the evidence on a small set of recording and elicitation conditions. Frequent use of EmoDB documents coverage of German, not validation of explanations across German speakers or recording environments, and a paper using several corpora contributes to several context

TABLE V
REPORTED TESTS OF THE EXPLANATION AND THE CLAIMS EACH CAN SUPPORT

Report	Test and observed result	What the test supports
[56]	Masking of individual SincNet filters; change in recognition confidence and accuracy.	Reliance on learned filters; not a complete perceptual explanation.
[54]	Removing high-attention segments of a speech–transcript model lowers accuracy by 2.44 percentage points, against 0.21 for low-attention segments.	The tested attention ranking; agreement with annotations is a separate matter from user benefit.
[49]	Insertion AUC and repeated-run top-feature Jaccard overlap on CREMA-D.	Decision recovery and ranking repeatability for this SER experiment only.
[50]	Less influential embedding components are replaced with audio-language-model output; decisions are compared with random and zeroed baselines.	Decision preservation under reconstruction; word error rate measures intelligibility.
[78]	Cue-saliency ablation and controlled cue perturbation, with a 16-participant user study.	Model sensitivity and listener performance under the tested conditions.
[33]	Saliency ablation; listeners compare matched real recordings with acoustic cues.	Listener benefit concerns real matched contrasts, not synthetic audio.
[18]	Concept vectors for voice pitch and utterance polarity retrained 30 times and compared with 50 random concepts by t-test.	That the model is sensitive to the concept direction; not that the concept causes the decision.
[19]	Two evaluations with ten raters each on acceptance of acoustic and content rationales, plus an LLM judge.	Perceived plausibility; not causal acoustic reliance.
[72]	Accuracy of acoustic descriptors and four explanation evaluations with 20 volunteers.	Separates the accuracy of cue descriptions from preference and perceived adequacy.
[20]	An LLM judge scores whether quoted words occur in the transcript (groundedness) and support the label (relevance).	That cited words exist and fit the label; not that they drove the decision.
[35]	The frequency profile of an utterance is replaced by the class-average profile of another class; the share of predictions moving to that class is reported.	That class-level frequency profiles drive the CNN decision on this corpus; not which time region of an utterance matters.

categories even when its explanation is illustrated on one example only. Weak recognition protocols also limit what an explanation can show. Augmentation before the train–test split and inconsistent sample totals [81], or insufficiently described splits [85], do not remove an explanatory figure, but they restrict what it says about generalisation; adding a saliency plot does not turn a weak recognition test into a strong explanation experiment.

F. Cross-Dimensional Synthesis

Taken together, the dimensions show an association between model access, explanatory unit and the question a method can answer. Classical models operating on descriptors mostly yielded aggregated feature rankings, convolutional and attention-based pipelines more often offered regions or frames, and language and audio-language models produced cue descriptions and quotations. Lexical evidence was explained by generated rationales in 4 of the 10 studies and by attribution, attention or concept tests in the others; none of these separated a word from its delivery. These are reporting patterns rather than limits of the methods: a descriptor model could be explained over sliding windows, and a neural model could be summarised globally.

Three implications follow. First, model complexity and explanatory detail do not move together. A global importance plot from a classical classifier can expose named acoustic variables more directly than the unnamed latent coordinates of a larger model, although naming a variable does not make its attribution faithful. Second, the claim has to match the evaluation. A saliency-ablation result supports a claim about model sensitivity, repeated-run overlap supports repeatability, agreement with annotations supports alignment, and an improvement in listeners' decisions supports utility for that task. Table V shows that these forms of evidence come from different reports and cannot be combined into a general statement that a method or architecture is validated; in particular, the few generative and contrastive reports with tests do not show that generation is superior, because their tasks, comparators and formats differ from those of the descriptive attribution reports. Third, the data context constrains all four dimensions. A stable ranking on one acted corpus may reflect its recording and elicitation conditions as much as emotion. Cross-corpus comparisons of band importance and descriptor relevance [76], [85], [91], [104] and the explanation-transfer correlations in FedEmoNet [61] are useful starting points, but they are not controlled tests of repeatability or fidelity. The recognition scores and explanation metrics are too heterogeneous for a pooled estimate, so we synthesise methodological patterns and evidential limits rather than rank methods by effectiveness.

G. Sensitivity Analyses

Table VI shows that the main patterns did not depend on borderline studies or on the speech information explained. In every subset, SHAP was the most frequent method, post-hoc analysis the most frequent relationship and aggregated explanations more common than localised ones. In the core set of 57 studies, 22 used SHAP, 35 gave aggregated explanations and only 8 tested the explanation, 4 of them for decision fidelity. Excluding the cross-domain studies removed two of the seven fidelity tests, so the evaluation gap is, if anything, wider among studies that focus on speech. Restricting the analysis to explanations of acoustic evidence alone left 22 SHAP studies and 7 with a test of the explanation, 6 of them for decision fidelity, so including lexical evidence did not change the main patterns. Raising the quality threshold to 5 removed five studies, none of which tested their explanations; the remaining 56 studies showed the same patterns, with 22 using SHAP, 35 post-hoc analyses, 33 aggregated and 19 localised explanations, and all 11 studies with a test of the explanation retained.

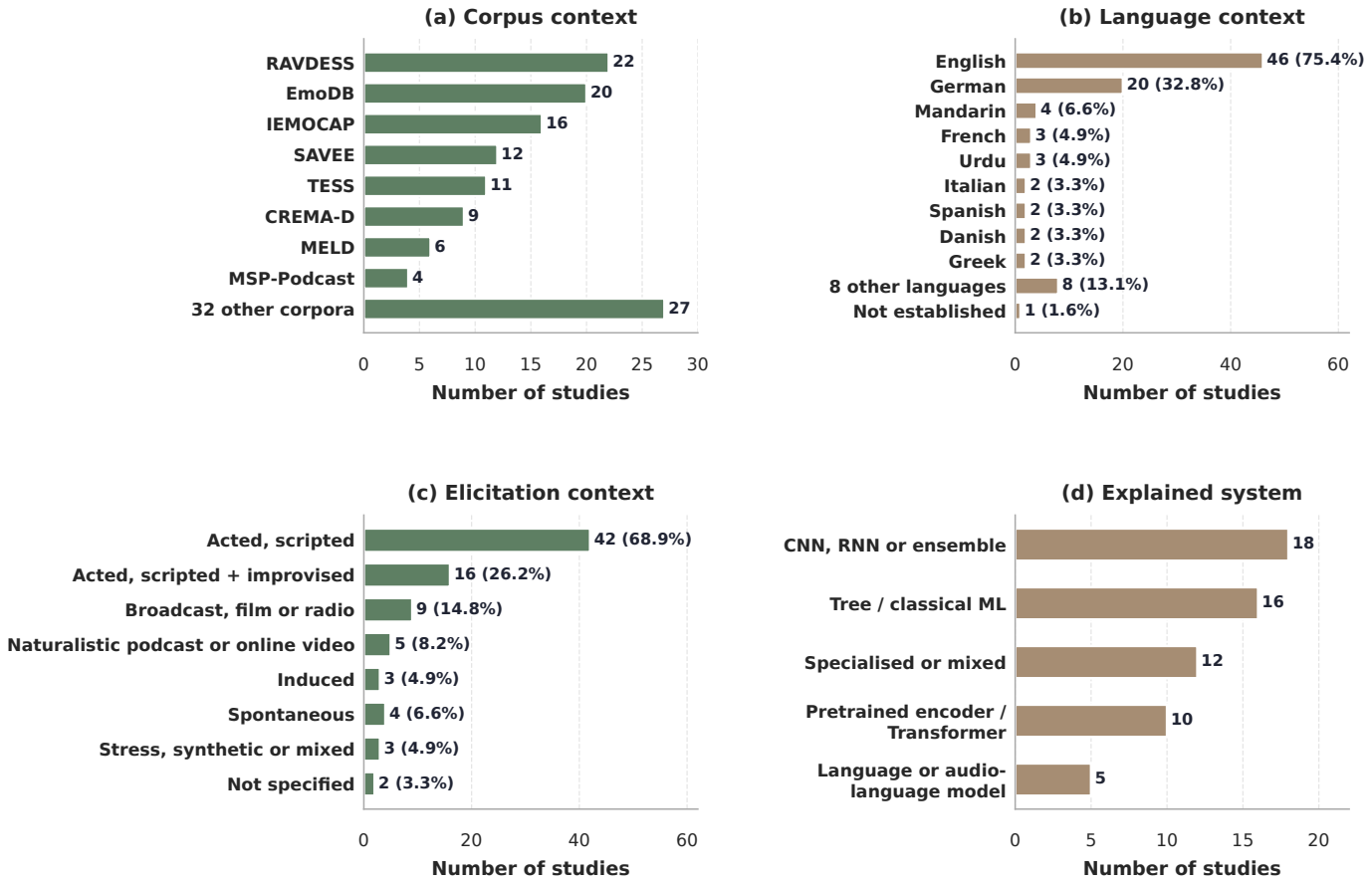


Fig. 10. Data and model contexts of the included studies: corpora in the evaluated pipeline (a), corpus language (b) and elicitation (c), both established from the corpus documentation, and type of explained system (d). A study can use several corpora, so (a)–(c) overlap; less frequent corpora and languages are grouped.

TABLE VI
SENSITIVITY OF THE MAIN PATTERNS TO BORDERLINE STUDIES AND TO THE SPEECH INFORMATION EXPLAINED. CELLS GIVE NUMBERS OF STUDIES. SHAP IS THE MOST FREQUENT METHOD IN EVERY SUBSET.

Subset	n	SHAP	Post-hoc (D1)	Aggregated (D2)	Localised (D2)	Acoustic descriptors (D3)	Decision fidelity	Any explanation test
All included studies	61	24	40	35	19	16	7	11
Acoustic evidence only	51	22	36	29	15	16	6	7
Excluding unclear RAVDESS subsets	59	23	39	35	18	16	6	10
Excluding cross-domain audio studies	59	23	38	35	17	16	5	9
Core set (both exclusions)	57	22	37	35	16	16	4	8
Stricter QA threshold (total at least 5)	56	22	35	33	19	15	7	11

VI. DISCUSSION

A. From a numerical target to an acoustic explanation

The central interpretive problem is the distance between what a model consumes, what an explainer attributes and what a listener can understand. The reviewed pipelines often pass through several representations, such as waveform, frame descriptors, pooled statistics, reduced coordinates and a learned state, and an attribution at one stage does not explain the stages before it. The SincNet studies interpret learned filters rather than waveform samples [56], [57], the feature-boosting reports explain principal components and relate them to descriptors afterwards [12], [39], and topological attribution concerns persistence-image coordinates [58]. Such outputs can be informative, provided the chain of transformations is preserved. The six studies whose explained target is unnamed indices show the most basic reporting gap: without a mapping, a reader cannot tell whether a highly ranked feature is a time point, a frequency bin or a selected latent coordinate [37], [47], [59]–[62].

This changes what should count as progress. A higher-resolution heatmap does not necessarily make an explanation

more useful, and an aggregated ranking is not inherently inadequate. A developer looking for a corpus shortcut may need a stable global summary; a listener checking one disputed label may need an interval and an audible contrast. Explanation quality should therefore be judged against a stated question and user, not resolution alone. Mapping relevance to acoustic concepts, or to words, is a constructive next step, but it adds a claim that also needs checking. Two recent preprints, outside the synthesis, point in this direction: one predicts interpretable acoustic descriptors from self-supervised speech embeddings to describe what those embeddings encode [31], and the other relates saliency regions to acoustic cues defined by experts [32]. Overlays of Grad-CAM with energy, fundamental frequency and formant trajectories connect model coordinates with speech description [83], [84], yet they show correspondence in the displayed example, not that changing pitch or a formant alone would alter the decision. A stronger study would document the mapping, quantify the correspondence on held-out utterances and test controlled cue changes while monitoring other acoustic properties.

Direction and magnitude deserve the same care. A rank says which feature receives more attribution under a given procedure, not whether a larger physical value raises or lowers a class score, and a global mean of absolute relevance discards the sign needed to interpret an individual decision. A statement such as "pitch is important" hides several questions: important for which emotion, for which examples, relative to which reference, and in which direction? These qualifiers should be kept whenever feature-level findings are translated into acoustic prose. They matter especially in corpus comparisons, where a feature can occupy the same rank in two models with different class-specific directions, and a change of rank can reflect changing correlations among descriptors rather than a change in the cue itself. The cross-corpus reports [61], [91], [104] motivate such comparisons, but the current evidence does not support a universal ordering of acoustic cues.

B. Faithfulness requires a specified decision and a matched control

The evaluation gap is less a shortage of metrics than a mismatch between the claim made about an explanation and the evidence collected for it. Recognition accuracy asks whether an emotion label is correct, faithfulness whether the purported evidence reflects the model's computation, and plausibility whether the explanation makes sense to a person. None of these implies the others [10], [11]: a correct label can come with an unsupported rationale, an understandable rationale can be unrelated to what the model relies on, and an accurate description of a model's behaviour can expose a spurious cue.

The qualifying tests probe reliance in different ways (Table V). Filter masking tests dependence on an intermediate representation [56]; removing high- versus low-attention segments compares an internal ranking with a control [54]; insertion AUC measures how quickly a decision is recovered as attributed evidence is restored [49]; regenerated audio tests whether a decision survives replacement of less influential components [50]; and saliency ablation tests the relevance of selected regions in cue-based systems [33], [78]. These scores are not interchangeable estimates of one property. Preserving a decision after reconstruction suggests that the retained evidence is sufficient under that procedure, whereas a larger drop for high-ranked than for low-ranked evidence indicates sensitivity to the selected evidence. Redundant cues can keep a decision unchanged after one relevant cue is removed, and reconstruction can introduce information absent from the original subset, so the competing explanations for an effect need to be stated alongside the manipulation.

Controls are particularly important for speech. Zeroing a spectral patch, replacing a latent component or masking a feature can create an unfamiliar input, so part of a score drop may reflect the distribution shift itself [105]. High-ranked evidence should therefore be compared with matched random or low-ranked evidence, the replacement procedure documented, and the manipulated audio checked for suitability. A claim about a fixed model requires perturbing that model; retraining after feature removal asks whether a new model can recover performance, which is useful for feature selection but answers a different question. Model- and data-randomisation checks, which test whether an attribution changes when learned information is destroyed [106], were absent from the reviewed evaluations. Reports should also state which outcome was measured, because confidence change for a specified class, label preservation and aggregate accuracy capture different responses: an explanation may change a class score without changing the winning label, and a class-average difference may hide variation between correctly and incorrectly classified items. Terms such as "faithful" or "causal" should be reserved for the scope these tests actually support.

C. Stability, transfer and explainer agreement answer different questions

Stability is easy to overstate because several superficially similar comparisons appear in the corpus. Repeated-run overlap asks whether a method produces similar rankings under the same setting [49]; controlled perturbation asks how explanations or estimated cues respond to a defined input change [78]; a cross-corpus comparison asks whether the model and its explanation behave similarly on other data [61], [91]; and correlation between SHAP and LIME asks whether two procedures agree on selected cases [61], [73]. The two SHAP–LIME correlations reported above illustrate the problem: they come from different models, targets, cases and statistics, so they can be neither compared nor pooled. Two methods may agree because they share a feature representation or reference, and two valid methods may disagree because they answer different local questions.

Nor should stability mean that an explanation never changes. If a modification alters an emotionally relevant cue and the decision changes, an unchanged explanation would be a problem. A useful protocol distinguishes nuisance changes, under which consistency is expected, from emotion-relevant changes, under which sensitivity is expected. For SER, this means stating the intended invariance, repeating explanations across seeds or sampling runs, separating within-speaker variation

from shifts in speaker, corpus and language, and fixing in advance the level at which agreement is measured. Repeated selection of a broad feature family is a weaker requirement than repeated selection of the same descriptor, and a stable global ranking can coexist with unstable local explanations; the top-feature overlap reported on CREMA-D [49] is evidence about that feature-set comparison, not about time-localised explanations. Transfer should also be split into recogniser and explainer: keeping the model fixed while changing the corpus differs from retraining and then comparing explanations, which changes both the decision function and the data.

D. Generated rationales need grounding and decision linkage

Verbal explanations offer an accessible interface to speech evidence, but fluent language can hide a weak link between the recording and the decision. The included systems address different parts of this problem. The speech language model produces acoustic and content rationales distilled from a teacher that was given the gold emotion label [19], and the chain-of-thought student is trained on rationales written by a larger model [74]; both can learn plausible descriptions associated with a label, but neither shows that a stated cue drove the student's decision. The captioning study additionally evaluates its pitch, volume and rate descriptors and collects human judgements [72], which provides grounding evidence, although accurate description and preference still do not test reliance. In the multitask audio-language model, a language-model judge found most quoted words in the transcript and fewer of them relevant to the label [20]; this checks that the rationale refers to the input, not that the model used it. Reinforcement learning with a format reward makes the rationale follow a fixed structure of content, acoustics and summary [40], which helps readers but adds no evidence about the decision.

Three claims should therefore be tested separately: that the event or word named in a rationale is present in the recording at an identifiable location, that it is relevant to the model's particular output rather than merely typical of the labelled emotion, and that communicating it helps the intended user. A rationale can satisfy one claim while failing another. Generated cue statements could be compared with measured contours or annotated events, quoted words with the transcript and its alignment, the cited evidence could be manipulated in a controlled way, and users' decisions could be compared with and without the explanation. General findings on unfaithful language-model explanations and hallucinated content support this caution [107], [108], but they cannot replace tests on speech. For the same reason, the audio-language-model reconstruction approach [50] should not be described as a natural-language rationale merely because it uses a generative model; its output is audio, and its test is decision agreement after reconstruction.

E. What is said and how it is said should be explained together

Lexical and acoustic evidence often point the same way, as when angry words are spoken loudly, but SER exists partly because they do not always do so: irony, politeness and suppressed emotion separate the words from their delivery, and these are the cases in which a user most needs to know which channel the model followed. The reviewed explanations do not yet answer that question. Rankings that mix transcript and audio features report modality shares that depend on how many features each modality contributes [62]; concept tests in separate unimodal branches show sensitivity within each branch but not their weight in the fused decision [18]; SHAP interactions on a surrogate describe the surrogate [71]; occluding the audio of a word removes both its meaning and its prosody [17]; token and waveform attributions computed side by side leave their relative weight open [36]; and rationales that name both a word and a vocal cue leave their relative influence open [19], [20], [40]. One subgroup analysis comes closer: valence was estimated far less accurately for utterances that contained a swear word but were rated positive (CCC 0.20, against 0.64 for the other utterances), which the authors attribute to verbal irony [17]. The result shows where the two channels diverge, but it does not show which of them the model followed.

Two further preprints, also outside the synthesis, show what such tests can reveal. A benchmark that crosses neutral and emotional wording with neutral and emotional prosody found that audio-language models largely followed the words: with lexically neutral sentences they mostly answered "neutral", and with conflicting cues they detected the conflict but rarely recovered the intended emotion [109]. Conversely, appending acoustic-cue tokens derived from eGeMAPS to the prompt of an audio-language model, and then shuffling, corrupting or contradicting them while the audio stayed fixed, showed that the model used the cues but remained partly anchored to the audio [110]. Neither study explains individual decisions, but both test which channel a model relies on, which none of the included studies did. Separating the channels requires interventions that change one while holding the other fixed. The transcript can be replaced by an ASR hypothesis, a neutral paraphrase or the empty string with the audio unchanged; the audio can be resynthesised with the same words and a different prosody, or replaced by a neutral reading of the same transcript; and attribution can be computed per modality against a stated reference and compared with the change in the decision when that modality is removed. Corpora with matched lexical content, such as acted sentences spoken in several emotions, already allow part of this design, and conversational corpora with mismatched cues allow the rest. Reports should also state whether the transcript was manual or recognised, since ASR errors change what the lexical explanation can refer to.

F. Audible explanations: useful contrasts rather than more displays

Speech offers the opportunity to explain in the same modality as the input, and an audible example can carry both what is said and how it is said. The audible outputs in the corpus range from salient regions converted back to audio [14] and regenerated decision-preserving speech [50] to cue-based counterfactual or contrastive examples [33], [78]. Sonification can also encode explanatory information for listening [23], but

mapping an attribution score to sound is an auditory encoding of the explanation, not a replay of the attributed segment, and its scale must convey the intended quantity without suggesting that the model heard the sonified signal. These forms answer different questions, such as where evidence occurs, what an alternative sounds like or how a cue differs, and their availability alone does not show that listeners understand the model better.

The RexNet study is informative precisely because it separates these conditions [33]. Its model experiments evaluate synthetic counterfactuals, whereas its user experiments use real matched recordings; the reported benefit of examples with cue comparisons therefore concerns that presentation and cannot be cited as evidence that synthetic speech improved users' decisions. Because the samples were selected for correct recognition and coherent explanation, the study also does not show how the interface behaves on errors, ambiguous utterances or unselected data. A naturally recorded contrast is easy to understand but differs in several acoustic properties at once, while a synthetic contrast can target one property but may also alter timbre, naturalness or intelligibility. Together with the smaller RobustRexNet study [78], this work provides a concrete starting point for evaluation. Future comparisons should keep a clear baseline, separate cue information from extra exposure to the recording, include erroneous or uncertain model outputs, and check synthetic contrasts for intelligibility, speaker identity and unintended cue changes. A richer interface may increase effort without improving decisions, so format combinations should be evaluated as experimental conditions.

G. Personalisation should be evaluated through the user's task

None of the included studies adapted explanations to an individual listener, so personalisation remains a research direction. The intended user and decision should be specified before a display is designed. A developer comparing model failures may benefit from feature distributions and reproducible interventions; a practitioner checking a disputed recording may need a short segment and a cue comparison; a person receiving feedback may need a plain-language description with the option to listen. User-centred XAI guidance emphasises matching explanations to users and tasks [11], [24]. For SER, a study could compare fixed and adapted explanations while measuring emotion decisions, error detection, time, workload and appropriate reliance. Acceptance and preference should be kept as separate outcomes, since an explanation may be liked without improving decisions, or improve them at the cost of effort, and model errors must be included to distinguish calibrated reliance from mere agreement with the system.

Personalisation also needs a stable evidential basis: changing terminology, detail or format should preserve the supported claim rather than produce a more persuasive but less grounded account. Explanation evaluation and interface evaluation should be kept apart, with the explanatory evidence established first and the way it is communicated compared afterwards. What the control group receives matters as well. If one condition offers a second recording, a cue description and more listening time, any improvement belongs to that combination unless the components are disentangled, and the included listener studies should be interpreted at the level of the conditions they tested [33], [78]. Factorial or carefully matched designs could separate the contributions of replay, contrastive example, visual map and acoustic prose.

H. Dataset and model context constrain generalisation

The recurrence of a few acted corpora provides a shared evidence base but also a shared limitation, because repeated use of the same datasets is not independent confirmation that a cue holds in spontaneous speech. Lexical explanations depend on corpora with meaningful and varied text, which excludes most acted corpora built on a few fixed sentences; they were therefore concentrated on IEMOCAP, on television dialogue and on English. EmoDB is acted German speech [93], IEMOCAP combines scripted and improvised acting [94], and MSP-Podcast contains naturalistic speech retrieved from podcast recordings [102]. Grouping these as "conversational", or leaving language unspecified, obscures differences that matter for interpretation.

The model context is equally important. In a pipeline with feature selection, dimensionality reduction, an encoder and an ensemble, an explanation of one component is not an explanation of the final system, and protocol problems carry over to explanations: if augmented copies of an utterance appear on both sides of a split, or test speakers overlap with training speakers, a convincing explanation on that test set cannot repair the evaluation. Transfer should be assessed at the explanation level as well as the recognition level, since cross-corpus accuracy does not reveal whether the model keeps the same cue dependence, and a changing attribution may be appropriate when the input distribution or the expression of emotion changes. Benchmark generalisation and emotional validity are also different questions. A model may reproduce corpus labels while learning cues tied to the recording protocol or expressive style, and this review concerns explanations of the model's decision, not whether that decision reveals a speaker's internal state. The clinical study in the corpus contributes an acoustic explanation for its specified emotional-speech task [103]; it is not evidence for diagnosis from general-purpose SER output. Where such properties cannot be recovered from a report, the uncertainty should be retained, as with the unclear RAVDESS subsets in our sensitivity analysis.

I. An evidence-driven research agenda

The preceding subsections examined each limitation in depth. Here they are condensed into six research directions, each stated as an open question with the minimum evidence that would answer it (Fig. 11), and into a reporting checklist (Table VII). The directions are ordered by dependence: an explanation must be traceable before its fidelity can be tested, and its fidelity should be established before its stability, its lexical and acoustic content, or its value to users can be interpreted.

TABLE VII
MINIMUM REPORTING ITEMS FOR XAI-SER STUDIES, DERIVED FROM THE GAPS IDENTIFIED IN THIS REVIEW.

#	Item	What to report
1	Explained predictor	Model and component explained, the output explained (class score, label or dimension) and the class or item selected
2	Data and protocol	Corpus, language, elicitation and subset; speaker-independent partition; transcript source (manual or ASR)
3	Explained target	Mapping from the input to the explained representation, with units (time, frequency band, named descriptor or word)
4	Method settings	Reference or baseline, perturbation scheme, background data, aggregation, and whether attributions are signed
5	Temporal resolution	Whether evidence is localised, and how it is aligned with time or words
6	Speech information	Whether the explanation concerns acoustic evidence, lexical evidence or both, and the contribution of each channel
7	Test of the explanation	Fidelity, stability or user test, with matched control, measured outcome, number of utterances and uncertainty
8	Selection of examples	How displayed examples were chosen, including errors and ambiguous items
9	Availability	Code, seeds and explanation outputs needed to reproduce the reported explanations

Direction 1: Traceable explanations. Six studies reported their explained target only as unnamed indices, and in many others the path from the recording to the explained coordinate was left implicit. *Open question:* to which acoustic quantity, time interval or word does each explained coordinate correspond, and in which direction does it move the score of the explained class? *Minimum evidence:* a documented mapping from input to explained target and signed, class-specific attributions (Section VI-VI-A).

Direction 2: Decision fidelity. Only 7 of the 61 studies tested whether the decision depended on the highlighted evidence, among them a single Shapley study, although Shapley values were the most frequent method. *Open question:* does altering the highlighted evidence change the decision of the same fixed model more than altering matched control evidence? *Minimum evidence:* a deletion, insertion, masking or regeneration test against random or low-ranked evidence, with the measured outcome stated (Section VI-VI-B). Existing tests [33], [35], [49], [50], [54], [56], [78] show that such designs are feasible for filters, spectrogram regions, latent components and attention.

Direction 3: Stability across runs, speakers, corpora and languages. Three studies tested the robustness or repeatability of their explanations, 32 used only acted, scripted corpora and 27 only English corpora. *Open question:* do explanations remain unchanged under nuisance variation and change under emotion-relevant variation, and do they transfer across speakers, corpora and languages when the model is held fixed? *Minimum evidence:* explanations repeated across seeds and evaluated under stated invariances, with recogniser transfer and explainer transfer reported separately (Sections VI-VI-C and VI-VI-H).

Direction 4: What is said versus how it is said. Ten studies explained lexical evidence, always together with acoustic evidence, and none determined which channel a given decision relied on. *Open question:* when words and delivery disagree, as in irony or suppressed emotion, which channel drives the decision, and does the explanation reflect it? *Minimum evidence:* interventions that change the transcript or the prosody while holding the other fixed, compared with per-channel attributions computed against a stated reference (Section VI-VI-E). Concept tests per branch [18], word occlusion [17] and joint audio–token attribution [36] provide starting points.

Direction 5: Grounded generated explanations. Six systems generated explanations by design, but only two scored their rationales, and these scores checked whether quoted words occurred in the input rather than whether they drove the decision [19], [20]. *Open question:* are the cues and words named in a rationale present in the recording, relevant to the particular decision, and consistent with the model's demonstrated reliance? *Minimum evidence:* comparison of stated cues with measured contours or aligned transcripts, followed by manipulation of the cited evidence (Section VI-VI-D).

Direction 6: User benefit of audible and personalised explanations. Four studies evaluated explanations with people, four produced audible output, and none adapted explanations to the listener. *Open question:* which format helps which user to make better decisions and to rely on the system appropriately, including when the model is wrong? *Minimum evidence:* task performance and calibrated reliance with and without explanations, on unselected items that include errors, with format components varied as separate conditions (Sections VI-VI-F and VI-VI-G). The listener studies built on cue-based contrasts [33], [78] show that such experiments can be run for SER.

Across all six directions, recognition performance, fidelity, stability and usability should be reported as separate outcomes, each with its own denominator and uncertainty. A new architecture, attribution technique or language interface can be a contribution in its own right, but any claim it makes about its explanations needs the corresponding test.

J. Limitations of this review

The publication window and the peer-reviewed-only criterion exclude work published before 2021 and relevant preprints, some of which are cited in the Discussion as supporting literature [31], [32], [109], [110], and partial coverage of 2026 limits temporal interpretation. We therefore do not claim exhaustive coverage.

Full-text decisions, quality scores and coding were not duplicated independently, so no inter-rater agreement can be reported, and one report [91] was assessed from the authors' version of the conference paper because the publisher version was not accessible. The taxonomy is an operational coding framework with overlapping method families, not a validated theory of explanation mechanisms. Borderline cases and reporting gaps remain visible through page locators and sensitivity analyses; the stability of the aggregate patterns under those analyses supports the descriptive conclusions but does not validate every individual coding decision. The unit

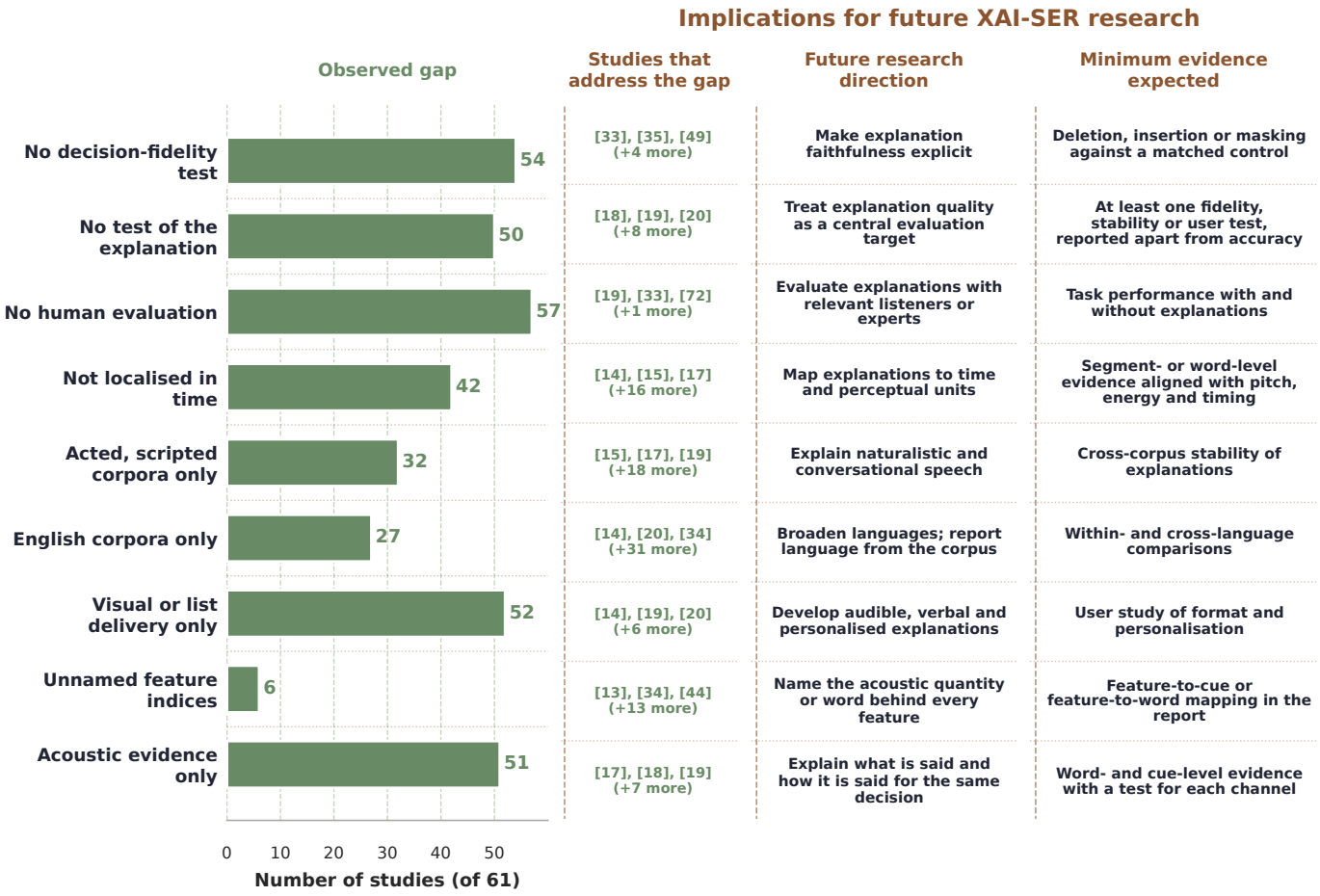


Fig. 11. Evidence gaps and research directions. Bars give the number of included studies showing each gap (gaps overlap); the second column cites studies that already address the gap.

of synthesis is the study, with one study reported twice and substantial reuse of benchmark corpora, so the evidence cannot support causal comparisons between explanation methods or a pooled effect estimate. Corpus–language mappings do not verify the subset used in each experiment. Systems that fuse speech with video were excluded unless the speech part of the explanation was reported separately, which removed several multimodal reports with substantial explanations. The lexical subset is small (10 studies), so the comparison between acoustic and lexical explanations is descriptive.

VII. Conclusion

This systematic review synthesised 61 studies (62 reports) and developed an operational, four-dimensional taxonomy of explanations for SER from speech and its transcript. Explanation methods differ not only in how they are computed but also in their relation to the trained recogniser, their temporal resolution, the representation they explain and whether they concern how something was said, what was said, or both. Post-hoc analysis was the most frequent approach (40 studies), SHAP the most common method (24), and aggregated evidence (35) was reported more often than evidence localised within an utterance (19). Lexical evidence was explained in 10 studies, always together with acoustic evidence, and four of them through generated rationales.

These distinctions determine what an explanation can establish. Attribution to acoustic descriptors can identify features associated with a model's decisions, whereas a spectrogram heatmap, a learned filter, an attention vector, a concept score and a generated rationale refer to different explanatory objects. A waveform input does not mean that the explanation is expressed in waveform time, a rationale about pitch or intensity does not show that those cues determined the label, and a quoted word does not show that the model relied on the word rather than on how it was spoken. Evidence that the explanations themselves are valid remains scarce: only 11 of the 61 studies tested the explanation, 7 for decision fidelity and 4 with human participants, and these tests addressed different questions that cannot be combined into a ranking of methods.

Future studies should specify the predictor, the emotion output, the explained representation and its mapping to perceptible acoustic cues or words; use matched interventions when testing reliance, including interventions that change the words and their delivery separately; and evaluate stability under stated conditions. Explanation quality, recognition performance and user benefit require separate measurements. Audible contrasts and personalised communication are promising

directions for speech-oriented XAI, but their value has to be shown in studies with representative users and with uncertain or incorrect predictions.

REFERENCES

[1] S. M. George and P. M. Ilyas, "A review on speech emotion recognition: A survey, recent advances, challenges, and the influence of noise," *Neurocomputing*, vol. 568, p. 127015, 2024. doi: 10.1016/j.neucom.2023.127015 .

[2] S. Al-Azani and E.-S. M. El-Alfy, "A review and critical analysis of multimodal datasets for emotional AI," *Artificial Intelligence Review*, vol. 58, p. 334, 2025. doi: 10.1007/s10462-025-11271-1 .

[3] S. Madanian, T. Chen, O. Adeleye, J. M. Templeton, C. Poellabauer, D. Parry, and S. L. Schneider, "Speech emotion recognition using machine learning—a systematic review," *Intelligent Systems with Applications*, vol. 20, p. 200266, 2023. doi: 10.1016/j.iswa.2023.200266 .

[4] S. Akinpelu, S. Viriri, and A. Adegun, "An enhanced speech emotion recognition using vision transformer," *Scientific Reports*, vol. 14, p. 13126, 2024. doi: 10.1038/s41598-024-63776-4 .

[5] W. Alsabhan, "Human–computer interaction with a real-time speech emotion recognition with ensembling techniques 1d convolution neural network and attention," *Sensors*, vol. 23, no. 3, p. 1386, 2023. doi: 10.3390/s23031386 .

[6] J. He, X. Shi, C.-H. Hu, J. Mi, X. Li, and T. Toda, "M4SER: Multimodal, multirepresentation, multitask, and multistrategy learning for speech emotion recognition," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 33, pp. 4055–4070, 2025. doi: 10.1109/taslpro.2025.3614428 .

[7] C. Luna-Jiménez, R. Kleinlein, D. Griol, Z. Callejas, J. M. Montero, and F. Fernández-Martínez, "A proposal for multimodal emotion recognition using aural transformers and action units on RAVDESS dataset," *Applied Sciences*, vol. 12, no. 1, p. 327, 2022. doi: 10.3390/app12010327 .

[8] Y. Wu, Q. Mi, and T. Gao, "A comprehensive review of multimodal emotion recognition: Techniques, challenges, and future directions," *Biomimetics*, vol. 10, no. 7, p. 418, 2025. doi: 10.3390/biomimetics10070418 .

[9] W. Saeed and C. Omlin, "Explainable AI (XAI): A systematic meta-survey of current challenges and future opportunities," *Knowledge-Based Systems*, vol. 263, p. 110273, 2023. doi: 10.1016/j.knosys.2023.110273 .

[10] M. Nauta, J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. Van Keulen, and C. Seifert, "From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable AI," *ACM Computing Surveys*, vol. 55, no. 13s, pp. 1–42, 2023. doi: 10.1145/3583558 .

[11] Y. Rong, T. Leemann, T.-T. Nguyen, L. Fiedler, P. Qian, V. Unhelkar, T. Seidel, G. Kasneci, and E. Kasneci, "Towards human-centered explainable AI: A survey of user studies for model explanations," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 4, pp. 2104–2122, 2024. doi: 10.1109/tpami.2023.3331846 .

[12] A. Nfissi, W. Bouachir, N. Bouguila, and B. L. Mishara, "Iterative feature boosting for explainable speech emotion recognition," in *Proc. 2023 International Conference on Machine Learning and Applications (ICMLA)*, 2023, pp. 543–549. doi: 10.1109/icmla58977.2023.00081 .

[13] S. I. M. M. Raton Mondol and S. Lee, "Explainability guided feature selection for speech emotion recognition using SHAP and permutation importance," *Biomedical Engineering Letters*, vol. 16, no. 5, pp. 1345–1356, 2026. doi: 10.1007/s13534-026-00592-x .

[14] T.-W. Kim and K.-C. Kwak, "Speech emotion recognition using deep learning transfer models and explainable techniques," *Applied Sciences*, vol. 14, no. 4, p. 1553, 2024. doi: 10.3390/app14041553 .

[15] T. Leygue, A. Sabourin, C. Bolzmacher, S. Bouchigny, M. Anastassova, and Q.-C. Pham, "Explainable speech emotion recognition through attentive pooling: Insights from attention-based temporal localization," in *Proc. Interspeech 2025*, 2025, pp. 4658–4662. doi: 10.21437/interspeech.2025-1841 .

[16] Y. Ma, X. Lu, J. Sang, X. Jiang, and R. Li, "Behind the scenes: Mechanistic interpretability of Lora-adapted Whisper for speech emotion recognition," in *Proc. ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 5286–5290. doi: 10.1109/icassp55912.2026.11464049 .

[17] A. Das, C. Franzreb, T. Polzehl, and S. Möller, "Exploring foundation model fusion effectiveness and explainability for stylistic analysis of emotional podcast data," in *Advances in Information and Communication*, ser. Lecture Notes in Networks and Systems, vol. 1283. Cham: Springer, 2025, pp. 370–386. doi: 10.1007/978-3-031-84457-$7_2$3 .

[18] A. R. Asokan, N. Kumar, A. V. Ragam, and S. S. Shylaja, "Interpretability for multimodal emotion recognition using concept activation vectors," in *Proc. Int. Joint Conf. Neural Networks (IJCNN)*, 2022. doi: 10.1109/IJCNN55064.2022.9892315 .

[19] B.-H. Su, H.-Y. Shih, J. Tian, J. Shi, C.-C. Lee, C. Busso, and S. Watanabe, "Reasoning beyond majority vote: An explainable Speechlm framework for speech emotion recognition," in *Proc. ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 22 532–22 536. doi: 10.1109/icassp55912.2026.11464563 .

[20] W. Zhang, Y. He, G. Lin, Z. Liu, S. Sun, B. Wang, X. Zou, J. H. M. Wong, Q. Wang, H. B. Sailor, N. F. Chen, and A. T. Aw, "Beyond classification: Towards speech emotion reasoning with multitask AudioLLMs," in *Proc. Int. Joint Conf. Natural Language Processing and Asia-Pacific Chapter of the ACL (IJCNLP-AACL)*, 2025. doi: 10.18653/v1/2025.ijcnlp-long.62 .

[21] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. García, S. Gil-López, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Information Fusion*, vol. 58, pp. 82–115, 2020. doi: 10.1016/j.inffus.2019.12.012 .

[22] A. Bibal, R. Cardon, D. Alfter, R. Wilkens, X. Wang, T. François, and P. Watrin, "Is attention explanation? an introduction to the debate," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Dublin, Ireland: Association for Computational Linguistics, 2022, pp. 3889–3900. doi: 10.18653/v1/2022.acl-long.269 .

[23] B. W. Schuller, T. Virtanen, M. Riveiro, G. Rizos, J. Han, A. Mesaros, and K. Drossos, "Towards sonification in multimodal and user-friendly explainable artificial intelligence," in *Proc. International Conference on Multimodal Interaction (ICMI)*, 2021, pp. 788–792. doi: 10.1145/3462244.3479879 .

[24] S. Hong and W. Park, "Developing user-centered system design guidelines for explainable AI: A systematic literature review," *Artificial Intelligence Review*, vol. 58, p. 386, 2025. doi: 10.1007/s10462-025-11363-y .

[25] T. Rathi and M. Tripathy, "Analyzing the influence of different speech data corpora and speech features on speech emotion recognition: A review," *Speech Communication*, vol. 162, p. 103102, 2024. doi: 10.1016/j.specom.2024.103102 .

[26] G. Alhussein, I. Ziogas, S. Saleem, and L. J. Hadjileontiadis, "Speech emotion recognition in conversations using artificial intelligence: A systematic review and meta-analysis," *Artificial Intelligence Review*, vol. 58, p. 198, 2025. doi: 10.1007/s10462-025-11197-8 .

[27] D. S. Johnson, O. Hakobyan, J. Paletschek, and H. Drimalla, "Explainable AI for audio and visual affective computing: A scoping review," *IEEE Transactions on Affective Computing*, vol. 16, no. 2, pp. 518–536, 2025. doi: 10.1109/taffc.2024.3505269 .

[28] H. M. Jayasinghe, K. W. Wong, and A. Nugaliyadde, "A systematic review of interpretability and explainability for speech emotion features in automatic speech emotion recognition," *Pattern Recognition*, vol. 171, p. 112122, 2026. doi: 10.1016/j.patcog.2025.112122 .

[29] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *PLOS Medicine*, vol. 18, no. 3, p. e1003583, 2021. doi: 10.1371/journal.pmed.1003583 .

[30] J. Priem, H. Piwowar, and R. Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts," arXiv preprint arXiv:2205.01833, 2022.

[31] S. Dixit, D. M. Low, G. Elbanna, F. Catania, and S. S. Ghosh, "Explaining deep learning embeddings for speech emotion recognition by predicting interpretable acoustic features," arXiv preprint arXiv:2409.09511, 2024.

[32] S. Nasr, Z. Ren, and D. Johnson, "Beyond saliency: Enhancing explanation of speech emotion recognition with expert-referenced acoustic cues," 2025, arXiv:2511.11691.

[33] W. Zhang and B. Y. Lim, "Towards relatable explainable AI with the perceptual process," in *Proc. CHI Conference on Human Factors in Computing Systems*. ACM, 2022, pp. 1–24. doi: 10.1145/3491102.3501826 .

[34] S. Das, N. N. Lønfeldt, A. K. Pagsberg, L. K. H. Clemmensen, and N. L. Lund, "Zero-shot cross-lingual speech emotion recognition: A study of loss functions and feature importance," in *Proc. 2nd Symp. Security and Privacy in Speech Communication (SPSC)*, 2022, pp. 23–29. doi: 10.21437/SPSC.2022-5 .

[35] M. de Velasco, R. Justo, A. López Zorrilla, and M. I. Torres, "Analysis of deep learning-based decision-making in an emotional spontaneous speech task," *Applied Sciences*, vol. 13, no. 2, p. 980, 2023. doi: 10.3390/app13020980 .

[36] A. Pandey, J. Singh, and M. Kaur, "Bridging text and speech for emotion understanding: An explainable multimodal transformer fusion framework with unified audio–text attribution," *Journal of Intelligence*, vol. 13, no. 12, p. 159, 2025. doi: 10.3390/jintelligence13120159 .

[37] M. Adeel, Z.-Y. Tao, S.-Y. Jin, C. Guo, and M. Alsuhaibani, "Assessing random forest performance in low resource speech emotion recognition," *Scientific Reports*, vol. 16, p. 854, 2025. doi: 10.1038/s41598-025-30511-6 .

[38] R. E. Donatus, "Interpretable speech emotion recognition: A comparative study of BiLSTM temporal attention and transformer-based multi-head self-attention," *Asian Journal of Electrical Sciences*, vol. 14, no. 2, pp. 21–27, 2025. doi: 10.70112/ajes-2025.14.2.4286 .

[39] A. Nfissi, W. Bouachir, N. Bouguila, and B. Mishara, "Unveiling hidden factors: explainable AI for feature boosting in speech emotion recognition," *Applied Intelligence*, vol. 54, no. 11-12, pp. 7046–7069, 2024. doi: 10.1007/s10489-024-05536-5 .

[40] P. Li, B. Zhao, Z. Kang, J. Peng, X. Qu, Y. He, and J. Wang, "EMO-RL: Emotion-rule-based reinforcement learning enhanced audio-language model for generalized speech emotion recognition," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. doi: 10.18653/v1/2025.findings-emnlp.1018 .

[41] S. Sharma, A. Dhingra, N. Bhanot, and S. Kharb, "Speech emotion recognition with explainable AI: Enhancing transparency in emotion recognition systems," in *Proc. 2025 4th OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 5.0*, 2025, pp. 1–6. doi: 10.1109/otcon65728.2025.11071221 .

[42] P. A. Amahan, "A retrieval-augmented generation method based on intent analysis for explainable speech emotion recognition," in *Proc. 2025 8th International Conference on Data Science and Information Technology (DSIT)*. IEEE, 2025, pp. 1–5. doi: 10.1109/dsit67006.2025.11389981 .

[43] N. Sripriya, S. Subramanian, and S. Jagathisan, "Multimodal emotional analysis using XAI for psychotherapy," *International Journal of Innovative Science and Research Technology*, vol. 10, no. 5, 2025. doi: 10.38124/ijisrt/25may474 .

[44] R. P, R. K. Dhaman, and A. Poulose, "Feature importance and model performance in deep learning for speech emotion recognition," in *Proc. 2024 11th International Conference on Advances in Computing and Communications (ICACC)*. IEEE, 2024, pp. 1–6. doi: 10.1109/icacc63692.2024.10845528 .

[45] R. K and B. R. Naik, "An interpretable hybrid deep learning framework for speech emotion recognition using self-supervised representations and bidirectional sequence modelling," *Journal of Intelligent Decision Making and Information Science*, vol. 3, no. 1s, pp. 348–368, 2026. doi: 10.59543/jidmis.v3.443 .

[46] A. Jain, L. Prasad, R. Yadav, and D. K. Yadav, "A deformable convolution and attention-driven multiscale transformer framework for robust and explainable speech emotion recognition," *Computer Methods in Biomechanics and Biomedical Engineering*, pp. 1–20, 2026. doi: 10.1080/10255842.2026.2667299 ., online first.

[47] M. Norval and Z. Wang, "Explainable artificial intelligence techniques for speech emotion recognition: A focus on XAI models," *Inteligencia Artificial*, vol. 28, no. 76, pp. 85–123, 2025. doi: 10.4114/intartif.vol28iss76pp85-123 .

[48] M. Mathivanan, S. Revathi, V. Porkodi, and B. Kasi, "Intelligent explainable dual path feature extraction and pyramid squeeze attention convolution network for speech emotion recognition," *Mobile Networks and Applications*, 2026. doi: 10.1007/s11036-026-02531-7 ., online first.

[49] S. Chakrabarty, P. Bishwas, M. Bandyopadhyay, and J. Sublime, "Can we trust AI with our ears? a cross-domain comparative analysis of explainability in audio intelligence," *IEEE Access*, vol. 13, pp. 179 733–179 758, 2025. doi: 10.1109/access.2025.3622161 .

[50] A. Akman, Q. Sun, and B. W. Schuller, "Improving audio explanations using audio language models," *IEEE Signal Processing Letters*, vol. 32, pp. 741–745, 2025. doi: 10.1109/lsp.2025.3532218 .

[51] T. Speith, "A review of taxonomies of explainable artificial intelligence (XAI) methods," in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2022, pp. 2239–2250. doi: 10.1145/3531146.3534639 .

[52] P. Kumar, V. Kaushik, and B. Raman, "Towards the explainability of multimodal speech emotion recognition," in *Proc. Interspeech*, 2021, pp. 1748–1752. doi: 10.21437/Interspeech.2021-1718 .

[53] H. Jang, C. Kim, and E. Yang, "TIMING: Temporality-aware integrated gradients for time series explanation," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 26 877–26 895.

[54] K. V. V. Girish, S. Konjeti, and J. Vepa, "Interpretabilty of speech emotion recognition modelled using self-supervised speech and text pre-trained embeddings," in *Proc. Interspeech 2022*, 2022, pp. 4496–4500. doi: 10.21437/interspeech.2022-10685 .

[55] H. Tao, Z. Hu, S. Li, C. Zhu, P. Li, and Y. Xie, "An interpretable multi-level feature disentanglement algorithm for speech emotion recognition," *IEICE Trans. Inf. Syst.*, vol. E108.D, no. 7, pp. 841–844, 2025. doi: 10.1587/transinf.2024EDL8083 .

[56] A. Anand, S. Negi, and N. Narendra, "Filters know how you feel: Explaining intermediate speech emotion classification representations," in *Proc. Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2021, pp. 756–760.

[57] A. Hekmat and Y. Shekofteh, "Persian speech emotion recognition using multi-scale SincNet and cGAN-based ensemble learning," in *Proc. 2026 International Interdisciplinary Conference on Artificial Intelligence: Engineering, Health, Finance and Humanities (IICAI)*. IEEE, 2026, pp. 1–5. doi: 10.1109/iicai70155.2026.11620846 .

[58] M. B. Raskin, "The shape of emotion in speech: A topological data analysis of speech emotion recognition," *IEEE Access*, vol. 14, pp. 52 964–52 978, 2026. doi: 10.1109/access.2026.3681238 .

[59] S. Yayavaram, A. Yayavaram, J. Christopher, and V. Arunachalam, "Interpretable feature optimization for sadness recognition in speech emotion analysis," in *Proc. 2024 IEEE 12th International Conference on Intelligent Systems (IS)*, 2024, pp. 1–10. doi: 10.1109/is61756.2024.10705229 .

[60] N. T. Pham, S. D. Nguyen, V. S. T. Nguyen, B. N. H. Pham, and D. N. M. Dang, "Speech emotion recognition using overlapping sliding window and Shapley additive explainable deep neural network," *Journal of Information and Telecommunication*, vol. 7, no. 3, pp. 317–335, 2023. doi: 10.1080/24751839.2023.2187278 .

[61] M. Tawfik, R. A. Obeidat, S. Kamel, N. A. Aljarrah, H. H. Shehadeh, and A. Dalalah, "FedEmoNet: Privacy-preserving federated learning with TCN-Transformer fusion for cross-corpus speech emotion recognition," *PLOS ONE*, vol. 21, no. 5, p. e0342953, 2026. doi: 10.1371/journal.pone.0342953 .

[62] A. Khalane, R. Makwana, T. Shaikh, and A. Ullah, "Evaluating significant features in context-aware multimodal emotion recognition with XAI methods," *Expert Systems*, vol. 42, no. 1, p. e13403, 2025. doi: 10.1111/exsy.13403 .

[63] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.

[64] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why should I trust you?": Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144. doi: 10.1145/2939672.2939778 .

[65] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70, 2017, pp. 3319–3328.

[66] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in *Proc. 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70, 2017, pp. 3145–3153.

[67] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626. doi: 10.1109/iccv.2017.74 .

[68] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847. doi: 10.1109/wacv.2018.00097 .

[69] S. Bach, A. Binder, G. Montavon, F. Klauschen, K.-R. Müller, and W. Samek, "On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation," *PLOS ONE*, vol. 10, no. 7, p. e0130140, 2015. doi: 10.1371/journal.pone.0130140 .

[70] B. Kim, M. Wattenberg, J. Gilmer, C. Cai, J. Wexler, F. Viegas, and R. Sayres, "Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV)," in *Proc. 35th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 2668–2677.

[71] Y. Meng, H. Wang, Y. Sun, and A. Li, "Joint acoustic–semantic alignment network for advanced speech emotion recognition," *Speech Communication*, vol. 181, p. 103409, 2026. doi: 10.1016/j.specom.2026.103409 .

[72] Q. Li, Y. Gao, Y. Wen, Y. Zhou, Z. Lian, B. Liu, Z. Wen, J. Tao, and Y. Li, "Exploring the use of large language models and interpretable features for explainable speech emotion recognition," *IEEE Journal of Selected Topics in Signal Processing*, vol. 20, no. 1, pp. 32–46, 2026. doi: 10.1109/jstsp.2026.3652299 .

[73] A. Samir, R. Bahgat, R. M. Badry, and B. Hassan, "A systematic evaluation of acoustic representation and feature fusion strategies for Arabic speech emotion recognition," *International Journal of Speech Technology*, vol. 29, no. 4, p. 70, 2026. doi: 10.1007/s10772-026-10290-9 .

[74] J. Mai, X. Xing, Y. Li, and X. Xu, "Chain-of-thought distillation with fine-grained acoustic cues for speech emotion recognition," in *Proc. Interspeech*, 2025, pp. 5438–5442. doi: 10.21437/Interspeech.2025-1979 .

[75] H. Lian, C. Lu, H. Yang, Y. Zhao, S. Li, and Y. Zong, "An interpretable collaborative acoustic parameter modeling network for speech emotion recognition," *IEEE Transactions on Computational Social Systems*, vol. 13, no. 4, pp. 4264–4280, 2026. doi: 10.1109/tcss.2026.3692891 .

[76] Y. Ma and W. Wang, "MSFL: Explainable multitask-based shared feature learning for multilingual speech emotion recognition," *Applied Sciences*, vol. 12, no. 24, p. 12805, 2022. doi: 10.3390/app122412805 .

[77] E. Jing, Y. Liu, Y. Chai, J. Sun, S. Samtani, Y. Jiang, and Y. Qian, "A deep interpretable representation learning method for speech emotion recognition," *Information Processing & Management*, vol. 60, no. 6, p. 103501, 2023. doi: 10.1016/j.ipm.2023.103501 .

[78] H. S. Abichandani, W. Zhang, and B. Y. Lim, "Robust relatable explanations of machine learning with disentangled cue-specific saliency," in *Proc. 30th International Conference on Intelligent User Interfaces (IUI)*, 2025, pp. 1203–1231. doi: 10.1145/3708359.3712105 .

[79] T. Ozseven, "Explainable transformer-based speech emotion recognition using benchmark datasets," in *Proc. 2026 4th Cognitive Models and Artificial Intelligence Conference (AICCONF)*. IEEE, 2026, pp. 1–7. doi: 10.1109/aicconf69182.2026.11600714 .

[80] P. Powroznik, M. Skublewska-Paszkowska, K. Dziedzic, M. Barszcz, K. Chwaleba, W. Wach, and V. Nunavath, "Explainable multimodal hybrid vision transformers for emotional speech recognition," in *Proc. 28th European Conference on Artificial Intelligence (ECAI 2025)*, ser. Frontiers in Artificial Intelligence and Applications, 2025. doi: 10.3233/faia250823 .

[81] H. Hasnat, K. Akter, and N. Tabassum, "Enhancing speech emotion recognition with explainable AI and feature importance analysis," in *Proc. 2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, 2025, pp. 1–6. doi: 10.1109/ncim65934.2025.11160160 .

[82] S.-V. Echim, R.-A. Smădu, and D.-C. Cercel, "Benchmarking adversarial robustness in speech emotion recognition: Insights into low-resource Romanian and German languages," in *Proc. 27th European Conference on Artificial Intelligence (ECAI 2024)*, ser. Frontiers in Artificial Intelligence and Applications, 2024. doi: 10.3233/faia240774 .

[83] L. Smietanka and T. Maka, "Interpreting convolutional layers in DNN model based on time–frequency representation of emotional speech," *Journal of Artificial Intelligence and Soft Computing Research*, vol. 14, no. 1, pp. 5–23, 2024. doi: 10.2478/jaiscr-2024-0001 .

[84] P. Singh, M. Sahidullah, and G. Saha, "Modulation spectral features for speech emotion recognition using deep neural networks," *Speech Communication*, vol. 146, pp. 53–69, 2023. doi: 10.1016/j.specom.2022.11.005 .

[85] K. G. Devi, K. Likhitha, J. Akshaya, R. F. J. Gokul, and G. J. Lal, "Multi-lingual speech emotion recognition: Investigating similarities between English and German languages," in *Proc. 2024 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI)*, 2024, pp. 1–10. doi: 10.1109/accai61061.2024.10601715 .

[86] J. H. Chowdhury, S. Ramanna, and K. Kotecha, "Speech emotion recognition with light weight deep neural ensemble model using hand crafted features," *Scientific Reports*, vol. 15, no. 1, 2025. doi: 10.1038/s41598-025-95734-z .

[87] G. R. K, V. U, and P. K. S. Karthinathan, "Innovative feature fusion and XAI framework for robust Tamil speech emotion recognition," in *Proc. 2026 7th International Conference on Inventive Research in Computing Applications (ICIRCA)*. IEEE, 2026, pp. 983–989. doi: 10.1109/icirca69024.2026.11570588 .

[88] M. Adeel and Z.-Y. Tao, "Enhancing speech emotion recognition in Urdu using Bi-GRU networks: An in-depth analysis of acoustic features and model interpretability," in *Proc. 2024 IEEE International Conference on Industrial Technology (ICIT)*. IEEE, 2024, pp. 1–6. doi: 10.1109/icit58233.2024.10540745 .

[89] H. E. Yildirim and D. Iren, "Informative speech features based on emotion classes and gender in explainable speech emotion recognition," in *Proc. 2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, 2023, pp. 1–8. doi: 10.1109/aciiw59127.2023.10388158 .

[90] A. S. Shinde and V. V. Patil, "Effect of data augmentation, cross-validation methods in robustness of explainable speech based emotion recognition," *Traitement du Signal*, vol. 41, no. 3, pp. 1565–1574, 2024. doi: 10.18280/ts.410344 .

[91] M. J. Hjuler, L. H. Clemmensen, and S. Das, "Exploring local interpretable model-agnostic explanations for speech emotion recognition with distribution-shift," in *Proc. ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5. doi: 10.1109/icassp49660.2025.10889825 .

[92] S. R. Livingstone and F. A. Russo, "The Ryerson audio-visual database of emotional speech and song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," *PLOS ONE*, vol. 13, no. 5, p. e0196391, 2018. doi: 10.1371/journal.pone.0196391 .

[93] F. Burkhardt, A. Paeschke, M. Rolfes, W. F. Sendlmeier, and B. Weiss, "A database of German emotional speech," in *Proc. Interspeech*, 2005, pp. 1517–1520.

[94] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee, and S. S. Narayanan, "IEMOCAP: Interactive emotional dyadic motion capture database," *Language Resources and Evaluation*, vol. 42, no. 4, pp. 335–359, 2008. doi: 10.1007/s10579-008-9076-6 .

[95] S. Haq and P. J. B. Jackson, "Speaker-dependent audio-visual emotion recognition," in *Proc. International Conference on Auditory-Visual Speech Processing (AVSP)*, 2009, pp. 53–58.

[96] K. Dupuis and M. K. Pichora-Fuller, "Toronto emotional speech set (TESS)," University of Toronto, Psychology Department, 2010. doi: 10.5683/sp2/e8h2mf .

[97] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "CREMA-D: Crowd-sourced emotional multimodal actors dataset," *IEEE Transactions on Affective Computing*, vol. 5, no. 4, pp. 377–390, 2014. doi: 10.1109/taffc.2014.2336244 .

[98] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "MELD: A multimodal multi-party dataset for emotion recognition in conversations," in *Proc. 57th Annu. Meeting Assoc. Computational Linguistics (ACL)*, 2019, pp. 527–536. doi: 10.18653/v1/P19-1050 .

[99] M. El Seknedy and S. Fawzi, "Speech emotion recognition system for human interaction applications," in *Proc. 2021 Tenth International Conference on Intelligent Computing and Information Systems (ICICIS)*, 2021, pp. 361–368. doi: 10.1109/icicis52592.2021.9694246 .

[100] B. H. Prasetio and E. R. Widasari, "Acoustic correlates of affective prosody across emotions and stress levels: A phonetic analysis with interpretable AI-assisted modeling," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 2904–2913, 2026. doi: 10.1109/taslpro.2026.3693971 .

[101] E. A. Alkhamali, A. A. Allinjawi, R. B. Ashari, and M. Tawfik, "FedSER-XAI: PSO-optimized multi-stream cross-attention transformer with graph features for explainable federated speech emotion recognition," *Scientific Reports*, vol. 15, no. 1, p. 43375, 2025. doi: 10.1038/s41598-025-28686-z .

[102] R. Lotfian and C. Busso, "Building naturalistic emotionally balanced speech corpus by retrieving emotional speech from existing podcast recordings," *IEEE Transactions on Affective Computing*, vol. 10, no. 4, pp. 471–483, 2019. doi: 10.1109/taffc.2017.2736999 .

[103] L. Chenain, A.-C. Bachoud-Lévi, and C. Clavel, "Acoustic characterization of Huntington's disease emotional expression: An explainable AI approach," in *Proc. 2024 12th International Conference on Affective*

Computing and Intelligent Interaction Workshops and Demos (ACIIW), 2024, pp. 247–255. doi: 10.1109/aciiw63320.2024.00052 .

[104] D. Casali, V. Cesarini, E. M. Carloni, and G. Costantini, "Language-dependent and language-independent acoustic cues in speech emotion recognition," in *Proc. 2026 IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0 & IoT)*. IEEE, 2026, pp. 742–747. doi: 10.1109/metroind4.0iot69397.2026.11653136 .

[105] X. Zheng, F. Shirani, Z. Chen, C. Lin, W. Cheng, W. Guo, and D. Luo, "F-Fidelity: A robust framework for faithfulness evaluation of explainable AI," in *Proc. International Conference on Learning Representations (ICLR)*, 2025. doi: 10.48550/arxiv.2410.02970 .

[106] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 9505–9515.

[107] M. Turpin, J. Michael, E. Perez, and S. R. Bowman, "Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 74 952–74 965.

[108] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 248:1–248:38, 2023. doi: 10.1145/3571730 .

[109] J. Chen, Z. Guo, J. Chun, P. Wang, A. Perrault, and M. Elsner, "Do audio LLMs really LISTEN, or just transcribe? Measuring lexical vs. acoustic emotion cues reliance," arXiv preprint arXiv:2510.10444, 2025.

[110] I. Tsangko, A. Triantafyllopoulos, and B. W. Schuller, "Acoustic cue alignment in audio language models for speech emotion recognition," arXiv preprint arXiv:2606.07309, 2026.

[111] J. Lin, "Enhancing speech emotion recognition through domain-aware data augmentation and model explainability," in *Proc. 2025 10th International Conference on Computer and Information Processing Technology (ISCIPT)*, 2025, pp. 520–527. doi: 10.1109/iscipt67144.2025.11265207 .

[112] X. Wang, R. Hu, J. Xie, R. Tian, J. Guan, and H. Ding, "Exploring the acoustic-prosodic characteristics of emotional prosody in Mandarin disyllabic speech materials with machine-learning tools," *Speech Communication*, vol. 184, p. 103473, 2026. doi: 10.1016/j.specom.2026.103473 .

[113] S. Sultana and M. S. Rahman, "Acoustic feature analysis and optimization for Bangla speech emotion recognition," *Acoustical Science and Technology*, vol. 44, no. 3, pp. 157–166, 2023. doi: 10.1250/ast.44.157 .

[114] V. Sidharrth, A. Pranav, and M. Khanna, "Interpretable feature selection for speech emotion recognition in New Zealand English with grid search optimization for effective hyperparameter tuning," in *ICT: Applications and Social Interfaces*, ser. Lecture Notes in Networks and Systems, vol. 1873. Cham: Springer, 2026, pp. 181–191. doi: 10.1007/978-3-032-19675-$0_1$8 .

[115] P. Du, K. Wang, H. Cui, and X. Liu, "Feature selection based on causal analysis algorithm for speech emotion recognition," in *Proc. 16th Int. Conf. Advanced Computer Theory and Engineering (ICACTE)*, 2023, pp. 36–40. doi: 10.1109/ICACTE59887.2023.10335104 .

[116] J. G. K. K, G. K. A, H. P. B, P. V. R, and V. D, "Graph neural network-based speech emotion recognition for emotion-aware support," in *Proc. IEEE Madhya Pradesh Section Conf. (MPCON)*, 2026, pp. 116–122. doi: 10.1109/MPCON69668.2026.11508484 .

[117] Q. Hu, Y. Peng, and Z. Zheng, "A deep learning framework for gender sensitive speech emotion recognition based on MFCC feature selection and SHAP analysis," *Scientific Reports*, vol. 15, no. 1, p. 28569, 2025. doi: 10.1038/s41598-025-14016-w .

[118] S. Depuru, P. J. Reddy, S. Sirisala, K. Amala, A. B. Reddy, and P. Vinod, "Federated meta-learning and explainable AI for hybrid multimodal emotion recognition," Springer conference chapter, pp. 231–242, 2026. doi: 10.1007/978-3-032-14044-$9_1$9 .

[119] Y. Chen, X. Xie, H. Xu, M. Geng, G. Li, C. Deng, H. Wang, S. Hu, and X. Liu, "Towards LLM-empowered fine-grained speech descriptors for explainable emotion recognition," in *Proc. Interspeech*, 2025, pp. 4633–4637. doi: 10.21437/Interspeech.2025-1112 .

[120] H. Pourrostami, M. Nazari, and M. M. AlyanNezhadi, "An explainable MFCC-based deep learning framework for song emotion recognition," in *Proc. 11th Int. Conf. Signal Processing and Intelligent Systems (ICSPIS)*, 2025, pp. 158–164. doi: 10.1109/ICSPIS68676.2025.11551721 .

[121] A. Amjad, L.-C. Tai, and H.-T. Chang, "Multimodal emotion recognition using AGSA-optimized LSTM with explainable AI," *IEEE Trans. Affective Computing*, 2026. doi: 10.1109/TAFFC.2026.3709670 ., early access.

[122] M. S. G. Padma *et al.*, "Explainable AI for emotion-aware human-computer interaction systems," in *Proc. Int. Conf. Innovations in Cybersecurity and Data Science (ICICDS)*, 2026. doi: 10.1109/ICICDS70526.2026.11604693 .

[123] B. Arumugam, S. D. Bhattacharjee, and J. Yuan, "Multimodal attentive learning for real-time explainable emotion recognition in conversations," in *Proc. IEEE Int. Symp. Circuits and Systems (ISCAS)*, 2022. doi: 10.1109/ISCAS48785.2022.9938005 .

[124] R. Acevedo, D. Onetto *et al.*, "Contrastive fusion network for interpreting multimodal emotional cues within interactive theatre performances," in *Proc. Int. Conf. Electrical Engineering and Informatics (ICEEI)*, 2025. doi: 10.1109/ICEEI68459.2025.11330581 .

[125] A. Thampy S. and J. J. R. Angelina, "Multimodal sentiment and emotion detection using hybrid fusion of text, audio, and image with explainable AI integration," *Optical Memory and Neural Networks*, vol. 35, no. 2, pp. 294–317, 2026. doi: 10.3103/S1060992X25602817 .

[126] A. A. Alyoubi and B. A. Alyoubi, "Interpretable multimodal emotion recognition using optimized transformer model with SHAP-based transparency," *J. Supercomputing*, vol. 81, p. 1044, 2025. doi: 10.1007/s11227-025-07515-0 .

[127] S.-K. Lee and J.-H. Kim, "CAMER: Contribution-aware multimodal emotion recognition," in *Proc. Interspeech*, 2025. doi: 10.21437/Interspeech.2025-114 .

[128] J. Tong, S. Li, Z. Zhuang, J. Hu, and Y. Zou, "EmoSync: Multi-stage reasoning with multimodal large language models for fine-grained emotion recognition," in *Proc. 3rd Int. Workshop on Multimodal and Responsible Affective Computing (MRAC)*, 2025. doi: 10.1145/3746270.3760231 .

[129] W.-C. Lin, K. Sridhar, and C. Busso, "An interpretable deep mutual information curriculum metric for a robust and generalized speech emotion recognition system," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 5117–5130, 2024. doi: 10.1109/TASLP.2024.3507562 .

[130] P. Kumar, S. Malik, B. Raman, and X. Li, "VISTANet: VIsual spoken textual additive net for interpretable multimodal emotion recognition," *IEEE Trans. Affective Computing*, vol. 16, no. 4, pp. 2881–2893, 2025. doi: 10.1109/TAFFC.2025.3573180 .

APPENDIX A
SEARCH STRATEGY

TABLE VIII: Search blocks and source-specific queries. Database and archive searches were last run on 2 November 2026 and citation searches on 7 November 2026; records are counted before matching across sources, and citation-search records are citation links.

Block / source	Query	Records
#1 SER block	(("speech emotion recognition" OR "speech emotion classification" OR "speech emotion detection" OR "emotion recognition from speech" OR "emotion classification from speech" OR "speech-based emotion recognition" OR "emotional speech recognition" OR "vocal emotion recognition" OR "voice emotion recognition" OR "acoustic emotion recognition" OR "speech emotion analysis" OR "speech emotion" OR "emotional speech" OR "spoken emotion" OR "vocal emotion" OR "voice emotion" OR "affective speech" OR "speech affect" OR "emotion in speech" OR "speech emotion reasoning" OR "emotion recognition from spoken language" OR "multimodal speech emotion recognition" OR "multimodal speech emotion analysis" OR "emotion recognition from speech and text" OR "emotion recognition from audio and text") OR (("multimodal emotion recognition" OR "multimodal emotion analysis" OR "conversational emotion recognition" OR "emotion recognition in conversation") AND (speech OR spoken OR voice OR vocal OR acoustic OR transcript))	–
#2 XAI block	("explainable artificial intelligence" OR "explainable AI" OR XAI OR explain* OR interpretab* OR "model interpretation" OR "model explanation" OR "decision explanation" OR attribution OR "feature attribution" OR "feature importance" OR saliency OR SHAP OR LIME OR "integrated gradients" OR "Grad-CAM" OR "layer-wise relevance propagation" OR LRP OR occlusion OR perturbation OR probing OR counterfactual* OR "representation analysis" OR "sensitivity analysis" OR "attention analysis" OR "contrastive explanation" OR "contrastive saliency" OR "contrastive cues" OR "concept-based explanation" OR "example-based explanation" OR "prototype-based explanation" OR "representation probing" OR "representation visualization" OR "attention visualization" OR "perturbation analysis" OR "natural language explanation" OR "natural language rationale" OR "rationale generation" OR "explanation generation" OR "chain-of-thought" OR "reasoning trace" OR sonification OR "auditory explanation" OR "acoustic explanation" OR "audio-based explanation" OR "personalized explanation" OR "personalised explanation" OR "adaptive explanation" OR "user-centered explanation" OR "explanation usability")	–
Scopus	TITLE-ABS-KEY(#1 AND #2)	640
IEEE Xplore	(#1) AND (#2)	277
Web of Science	TS=(#1 AND #2)	305
ACM Digital Library	#1 and #2, each applied to the Title, Abstract and Keyword fields	46
ACL Anthology	Shortened versions of #1 and #2 (ten searches)	10
ISCA Archive	Shortened versions of #1 and #2 (five searches)	23
OpenAlex (backward)	filter=cited_by:*seed IDs*,title_and_abstract.search:emotion; three seeds per query, 20 queries; run 7 November 2026	1092
OpenAlex (forward)	filter=cites:*seed IDs*,title_and_abstract.search:emotion; three seeds per query, 20 queries; run 7 November 2026	358

APPENDIX B
CODING OF THE INCLUDED REPORTS

The appendices are organised by unit of record. Appendix A gives the search strings and citation-search queries; this appendix gives, for each included study, the extraction fields and the primary and secondary codes in one table; Appendix C gives the quality scores of every assessed report and the reasons for each exclusion. Study-level evidence that is too long for a table, namely the observed explanatory output with its page, figure or table, the outcome of each test of the explanation, the corpus language and elicitation, and study-specific limitations, is given in the Extraction and Taxonomy sheets of the supplementary data workbook, with the same identifiers.

Codes follow Section IV. D1: Post-hoc; Built-in; Design = generated by design; Combined = built-in and post-hoc; Guided = explanation-guided. D2: Loc = localised; Agg = aggregated; Unres = unresolved. Info. (speech information): A = acoustic; A+L = acoustic and lexical. Del. (delivery format): Vis = plot, map or table; List = ranked list or prose; Verb = verbal; Aud = includes audible output. Eval.: F = decision fidelity; S = robustness or stability; H = human evaluation; R = automated scoring of generated rationales; – = descriptive only.

TABLE IX: Explained system, corpora and taxonomy codes of the 61 included studies

Ref.	Year	Explained system	Corpora	Methods (D4)	D1	D2	D3	Info.	Del.	Eval.
[13]	2026	XGBoost; comparison with SVM and random forest	RAVDESS, SAVEE, TESS, CREMA-D	SHAP, Permutation importance	Guided	Agg	Descriptors	A	List	–
[14]	2024	YAMNet and VGGish transfer pipeline	AI-Hub, CSU 2021, CSU 2022	LIME, Grad-CAM, Occlusion	Post-hoc	Loc	T–F regions	A	Aud	–
[15]	2025	W2V-BERT 2.0 audio, DeBERTa-v3 text, attentive statistics pooling	MSP-Podcast	Attention analysis	Built-in	Loc	Frames/words	A	Vis	–
[16]	2026	LoRA-adapted Whisper-large-v2	IEMOCAP	Layer probing	Post-hoc	Agg	Learned states	A	Vis	–
[17]	2025	Late fusion of pretrained speech encoders	MSP-Podcast	Occlusion	Post-hoc	Loc	T–F regions	A+L	Vis	–
[18]	2022	BC-LSTM; TCAV on the unimodal audio and transcript bottlenecks	IEMOCAP	TCAV	Post-hoc	Agg	Concepts	A+L	Vis	S

Ref.	Year	Explained system	Corpora	Methods (D4)	D1	D2	D3	Info.	Del.	Eval.
[19]	2026	Qwen2-Audio-7B-Instruct student and Qwen2.5-Omni teacher	RAVDESS, IEMOCAP, CREMA-D, MELD, MSP-Podcast	Generated rationale	Design	Agg	Concepts	A+L	Verb	H,R
[20]	2025	Multitask AudioLLM trained on emotion-reasoning targets	IEMOCAP, MELD, M3ED	Generated rationale	Design	Agg	Concepts	A+L	Verb	R
[33]	2022	RexNet: mel-spectrogram CNN, StarGAN-VC and a cue-comparison module	RAVDESS	Grad-CAM, Counterfactual audio, Contrastive acoustic cues, LRP	Combined	Loc	Mixed	A	Aud	F,H
[34]	2022	VAE with a two-dimensional latent space (four loss variants) and a…	EmoDB, IEMOCAP, SAVEE, AESSD, CaFE, Urdu corpus	DeepLIFT	Post-hoc	Agg	Descriptors	A	Vis	–
[35]	2023	Small CNN (about 43K parameters) on spectrograms with categorical…	Spanish TV debate corpus	Class model visualisation	Post-hoc	Agg	T–F regions	A	Vis	F
[36]	2025	RoBERTa (transcript) and WavLM (audio) projected to a shared space…	MELD	Integrated Gradients, Occlusion	Post-hoc	Loc	Mixed	A+L	Vis	–
[37]	2025	Random forest on MFCC (with deltas and chroma in extended runs)	SEMOUR+	SHAP, Permutation importance, Tree importance	Post-hoc	Agg	Indices	A	Vis	–
[38]	2025	BiLSTM with temporal attention; one-block Transformer with…	RAVDESS	Attention analysis	Built-in	Loc	Frames/words	A	Vis	–
[12], [39]	2024	Extra Trees on boosted PCA features	RAVDESS, EmoDB, SAVEE, TESS	SHAP	Guided	Agg	Transformed	A	Vis	–
[40]	2025	Qwen2-Audio-7B-Instruct trained with GRPO and emotion rules (EMO-RL)	RAVDESS, IEMOCAP, SAVEE, MELD	Generated rationale	Design	Agg	Concepts	A+L	Verb	–
[44]	2024	Random forest and decision tree	TESS	Tree importance	Post-hoc	Agg	Descriptors	A	Vis	–
[45]	2026	Self-supervised representations and BiLSTM	TESS	Gradient attribution	Post-hoc	Unres	Mixed	A	Vis	–
[46]	2026	DUST-Net, deformable convolution and multiscale Transformer	RAVDESS, EmoDB, CREMA-D, ASVP-ESD	DeepLIFT	Post-hoc	Unres	T–F regions	A	Vis	–
[47]	2025	CNN and ConvLSTM	own Afrikaans	SHAP, LIME	Post-hoc	Unres	Indices	A	Vis	–
[48]	2026	Dual-path 1D/2D PSAConvNN	RAVDESS	SHAP, TCAV	Post-hoc	Unres	Mixed	A	Vis	–
[49]	2025	CNN with MFCC and spectrogram inputs	CREMA-D	SHAP, LIME, Grad-CAM	Post-hoc	Loc	T–F regions	A	Vis	F,S
[50]	2025	EnCodec embedding classifiers and AudioGen	TESS	Integrated Gradients, Generative audio	Post-hoc	Loc	Mixed	A	Aud	F
[54]	2022	Sub-utterance speech–text model (SSPM-ATT)	IEMOCAP	Attention analysis	Built-in	Loc	Frames/words	A+L	Vis	F
[56]	2021	SincNet-based emotion classifier	IEMOCAP	Learned filters	Built-in	Agg	Learned states	A	Vis	F
[57]	2026	One-vs-all multi-scale SincNet ensemble	ShEMO	Learned filters	Built-in	Agg	Learned states	A	Vis	–
[58]	2026	TDA MLP, CRNN and fusion	RAVDESS, SAVEE, TESS, CREMA-D	Integrated Gradients	Post-hoc	Agg	Learned states	A	Vis	–
[59]	2024	SVM and KNN evaluated; SHAP model linkage under-specified	RAVDESS, EmoDB, SAVEE, TESS	SHAP	Post-hoc	Unres	Indices	A	Vis	–
[60]	2023	Deep pyramidal residual network	EmoDB, AESSD	SHAP	Post-hoc	Agg	Indices	A	Vis	–
[61]	2026	Federated CNN/TCN/Transformer (FedEmoNet)	RAVDESS, EmoDB, CREMA-D	SHAP, LIME, Attention analysis	Combined	Loc	Indices	A	Vis	–
[62]	2025	Early-fusion BiLSTM; audio-text bimodal models (GloVe and BERT…	CMU-MOSEI	SHAP	Guided	Agg	Indices	A+L	List	–
[71]	2026	ASNet (CNN acoustic branch, BERT transcript branch, alignment);…	IEMOCAP, MELD	SHAP	Post-hoc	Agg	Transformed	A+L	Vis	–
[72]	2026	WavLM-Large, UniAdapter and LLaMA-2 (SEmoLLM)	RAVDESS, IEMOCAP, MSP-Podcast, MSP-IMPROV	Generated caption	Design	Agg	Concepts	A	Verb	H
[73]	2026	ExtraTrees and XGBoost base learners, voting ensemble, wav2vec2…	EYASE	SHAP, LIME, Grad-CAM	Post-hoc	Loc	Mixed	A	Vis	–
[74]	2025	AECoTD: Qwen2.5-7B student distilled from DeepSeek-R1 on…	IEMOCAP, MELD	Generated rationale	Design	Loc	Concepts	A+L	Verb	–

Ref.	Year	Explained system	Corpora	Methods (D4)	D1	D2	D3	Info.	Del.	Eval.
[75]	2026	ICAPM-Net	EmoDB, IEMOCAP, ABC	Network weights	Built-in	Agg	Learned states	A	Vis	–
[76]	2022	Multitask shared-feature network (MSFL)	EmoDB, SAVEE, CASIA	Attention analysis	Built-in	Agg	Descriptors	A	Vis	–
[77]	2023	Interpretable group CNN (IG-CNN)	RAVDESS, IEMOCAP, CREMA-D, eNTERFACE'05	Decision coefficients	Built-in	Agg	Learned states	A	Vis	–
[78]	2025	RobustRexNet	ESD	Counterfactual audio, Cue saliency	Design	Loc	Mixed	A	Aud	F,S,H
[79]	2026	wav2vec 2.0 and Transformer	RAVDESS, IEMOCAP	SHAP, Attention analysis, Occlusion	Combined	Loc	Frames/words	A	Vis	–
[80]	2025	EM-H-ViT on four acoustic transforms	EmoDB, SAVEE, Danish DB, Polish DB	Grad-CAM, Attention analysis	Combined	Loc	T–F regions	A	Vis	–
[81]	2025	1D CNN	RAVDESS, SAVEE, TESS, CREMA-D	LIME, Integrated Gradients	Post-hoc	Loc	Mixed	A	Vis	–
[82]	2024	AlexNet and convolutional vision Transformer	EmoDB, Emo-IIT	Grad-CAM++	Post-hoc	Loc	T–F regions	A	Vis	–
[83]	2024	CNNs on constant-Q time–frequency inputs	EmoDB	Grad-CAM	Post-hoc	Loc	T–F regions	A	Vis	–
[84]	2023	CNN front end with SVM back end	RAVDESS, EmoDB	Grad-CAM	Post-hoc	Loc	T–F regions	A	Vis	–
[85]	2024	CNN	RAVDESS, EmoDB	LIME	Post-hoc	Unres	T–F regions	A	Vis	–
[86]	2025	Averaged 1D CNN and CNN–BiLSTM ensemble	RAVDESS, EmoDB, SAVEE, TESS, CREMA-D	LIME	Post-hoc	Unres	Mixed	A	Vis	–
[87]	2026	Random forest and XGBoost	EmoTa	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–
[88]	2024	Bidirectional GRU	Urdu corpus	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–
[89]	2023	Multiple classical classifiers	EmoDB	SHAP	Post-hoc	Agg	Descriptors	A	List	–
[90]	2024	XGBoost	EmoDB	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–
[91]	2025	ComParE-SVC and wav2vec 2.0-SVC	RAVDESS, EmoDB, IEMOCAP	LIME	Post-hoc	Agg	T–F regions	A	Vis	–
[99]	2021	Logistic regression, SVM, random forest and MLP	RAVDESS, EmoDB, CaFE	Permutation importance	Post-hoc	Agg	Descriptors	A	List	–
[100]	2026	TDNN and SVM	RAVDESS, EmoDB, IEMOCAP, SAVEE, TESS, DEMoS, DES, EmoV-DB, SUSAS	SHAP, Permutation importance	Post-hoc	Agg	Descriptors	A	Vis	–
[101]	2025	Federated multi-stream cross-attention Transformer	EmoDB, SAVEE	SHAP, LIME	Post-hoc	Agg	Mixed	A	Vis	–
[103]	2024	Random forest on acoustics and transcript features	own French (HD)	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–
[104]	2026	Random forest and SVM compared	EmoFilm	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–
[111]	2025	XGBoost	TESS	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–
[112]	2026	SVM, random forest and MLP	own Mandarin	SHAP, Permutation importance	Post-hoc	Agg	Descriptors	A	List	–
[113]	2023	XGBoost	RAVDESS, SUBESCO	Tree importance	Post-hoc	Agg	Descriptors	A	Vis	–
[114]	2026	Random forest	JL Corpus	SHAP	Post-hoc	Agg	Descriptors	A	Vis	–

APPENDIX C
QUALITY ASSESSMENT AND FULL-TEXT DECISIONS

Each criterion is scored 0, 0.5 or 1 according to Table III: Q1 = task and data origin (emotion recognition from spoken speech, its transcript, or both); Q2 = data and preparation (corpus, classes, preparation and sampling); Q3 = evaluation protocol (partition, evaluation unit and test procedure); Q4 = explanation approach (a concrete method applied to an identified emotion decision or model component); Q5 = explanation evidence (a figure, table, concept score or rationale linked to emotion decisions); Q6 = recognition assessment (metrics on stated test data with comparisons); Q7 = interpretation and reliability (1 = a fidelity, stability, robustness, human or rationale-grounding test; 0.5 = interpretation without a test). Total = sum of Q1–Q7 (maximum 7). QA = Pass when the total is at least 4 and Q1, Q4 and Q5 are all above zero. Eligibility was decided first and the QA gate was applied only to eligible reports. Reports excluded at eligibility (X1–X6, Table II) were also scored on Q1–Q7 so that all assessed reports have scores; these scores are informative only, the QA gate was not applied to them (QA = Info.), and the decision column gives the eligibility exclusion (Excl. X1–X6). Below QA = eligible but excluded after quality assessment. Companion = conference version of an included study, scored but counted with that study.

TABLE X: QA v2 scores and full-text decisions for all 83 assessed reports (77 from database and archive searches, 6 from citation searching): 61 included studies, one companion report, 3 reports below the quality threshold and 18 reports excluded at eligibility, whose reasons are given below the table.

Ref.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total	QA	Decision
Included studies (61) and companion report (1)										
[13]	1	1	0.5	1	1	1	0.5	6	Pass	Include
[14]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[15]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[16]	1	1	1	1	0.5	1	0.5	6	Pass	Include
[17]	1	1	1	1	1	1	0.5	6.5	Pass	Include

Ref.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total	QA	Decision
[18]	1	1	1	1	1	0.5	1	6.5	Pass	Include
[19]	1	1	1	1	1	1	1	7	Pass	Include
[20]	1	1	1	1	1	1	1	7	Pass	Include
[33]	1	0.5	0.5	1	1	1	1	6	Pass	Include
[34]	1	1	0.5	1	1	1	0.5	6	Pass	Include
[35]	1	1	0.5	1	1	1	1	6.5	Pass	Include
[36]	1	1	0.5	1	1	0.5	0.5	5.5	Pass	Include
[37]	1	0.5	0.5	1	0.5	0.5	0	4	Pass	Include
[38]	1	1	0.5	1	1	0.5	0.5	5.5	Pass	Include
[39]	1	1	1	1	0.5	1	0.5	6	Pass	Include
[40]	1	0.5	0.5	1	0.5	1	0.5	5	Pass	Include
[44]	1	0.5	0	0.5	1	0.5	0.5	4	Pass	Include
[45]	1	1	0.5	0.5	0.5	0.5	0.5	4.5	Pass	Include
[46]	1	0.5	0.5	1	0.5	0.5	0.5	4.5	Pass	Include
[47]	1	0.5	0.5	1	0.5	0.5	0.5	4.5	Pass	Include
[48]	1	0.5	0.5	1	1	0.5	0.5	5	Pass	Include
[49]	1	1	1	1	1	0.5	1	6.5	Pass	Include
[50]	1	1	1	1	1	1	1	7	Pass	Include
[54]	1	1	1	1	1	1	1	7	Pass	Include
[56]	1	1	1	1	1	1	1	7	Pass	Include
[57]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[58]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[59]	1	1	0.5	1	0.5	0.5	0.5	5	Pass	Include
[60]	1	1	0.5	1	1	0.5	0.5	5.5	Pass	Include
[61]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[62]	1	1	0.5	1	1	1	0.5	6	Pass	Include
[71]	1	1	1	0.5	1	1	0.5	6	Pass	Include
[72]	1	1	1	1	1	1	1	7	Pass	Include
[73]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[74]	1	1	1	1	0.5	1	0.5	6	Pass	Include
[75]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[76]	1	1	1	0.5	0.5	1	0.5	5.5	Pass	Include
[77]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[78]	1	1	1	1	1	1	1	7	Pass	Include
[79]	1	1	0.5	1	1	0.5	0.5	5.5	Pass	Include
[80]	1	0.5	1	1	1	1	0.5	6	Pass	Include
[81]	1	0.5	0.5	1	1	0.5	0.5	5	Pass	Include
[82]	1	1	0.5	1	1	0.5	0.5	5.5	Pass	Include
[83]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[84]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[85]	1	0.5	0.5	1	1	0.5	0.5	5	Pass	Include
[86]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[87]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[88]	1	0.5	0.5	1	1	0.5	0.5	5	Pass	Include
[89]	1	1	1	1	0.5	1	0.5	6	Pass	Include
[90]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[91]	1	0.5	1	1	1	0	0.5	5	Pass	Include
[99]	1	1	1	1	0.5	1	0.5	6	Pass	Include
[100]	0.5	1	1	1	1	1	0.5	6	Pass	Include
[101]	1	1	1	1	1	0.5	0.5	6	Pass	Include
[103]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[104]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[111]	1	1	0.5	1	1	0.5	0.5	5.5	Pass	Include
[112]	1	1	1	1	1	1	0.5	6.5	Pass	Include
[113]	1	1	1	0.5	0.5	1	0	5	Pass	Include
[114]	1	1	0.5	1	1	1	0.5	6	Pass	Include
[12]	1	1	1	1	1	1	0.5	6.5	Pass	Companion
Eligible but below the QA threshold (3)										
[41]	1	0.5	0	0.5	0.5	0.5	0	3	Fail	Below QA
[42]	1	0.5	0	0.5	0.5	0.5	0	3	Fail	Below QA
[43]	1	0.5	0	0.5	0.5	0.5	0.5	3.5	Fail	Below QA
Excluded at eligibility (18); QA computed for information only										
[52]	1	1	1	0.5	0.5	1	0	5	Info.	Excl. X1
[55]	1	0.5	0.5	0.5	0	1	0	3.5	Info.	Excl. X1
[115]	1	0.5	0.5	0	0	0.5	0	2.5	Info.	Excl. X1
[116]	0.5	0.5	0.5	0.5	0	0.5	0	2.5	Info.	Excl. X2
[117]	1	0.5	0.5	1	0.5	0.5	0	4	Info.	Excl. X2
[118]	0.5	0.5	0	0.5	0	0.5	0	2	Info.	Excl. X2
[119]	1	1	1	0.5	0	1	0	4.5	Info.	Excl. X2
[120]	0	0.5	1	1	1	1	0.5	5	Info.	Excl. X3
[121]	0.5	1	0.5	0.5	0.5	1	0	4	Info.	Excl. X4
[122]	0.5	0.5	0.5	0.5	0.5	0.5	0	3	Info.	Excl. X4
[123]	0.5	1	0.5	0.5	0.5	1	0	4	Info.	Excl. X4
[124]	0.5	0.5	0	0.5	0.5	0.5	0	2.5	Info.	Excl. X4
[125]	0.5	1	0.5	0.5	0.5	1	0	4	Info.	Excl. X4

Ref.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total	QA	Decision
[126]	0.5	1	0.5	0.5	0.5	0.5	0	3.5	Info.	Excl. X4
[127]	0.5	1	0.5	0.5	0.5	1	0	4	Info.	Excl. X4
[128]	0.5	1	1	0.5	0	1	0	4	Info.	Excl. X4
[129]	1	1	1	0.5	0.5	1	0	5	Info.	Excl. X5
[130]	0.5	1	1	1	0.5	1	0.5	5.5	Info.	Excl. X6

Reasons for exclusion. [41] Q: QA v2 total 3/7, below the threshold of 4. A local explanation is shown for an identified angry test utterance. Anonymous input indices do not support claims about named acoustic cues. [42] Q: QA v2 total 3/7, below the threshold of 4. Table IV pairs predictions with retrieved WAV filenames and brief cue phrases for five emotions, but the cue phrases are truncated and class-level, and their relationship to individual query recordings is insufficiently documented. [43] Q: QA v2 total 3.5/7, below the threshold of 4. LIME is applied to audio spectrograms, but the explained model is not specified and no evaluation protocol is reported. [52] X1: Embedding projection and class intersection matrices assess joint-representation separability, but do not connect identifiable acoustic or spoken linguistic evidence to emotion decisions. Audio–text setting itself is eligible. [55] X1: Two-dimensional cluster plot (Fig. 2) and accuracy ablation only; no cue, component or pathway linked to a decision. [115] X1: Causal feature selection; figures report selected-feature counts and recognition rates. No feature is linked to an emotion decision. [116] X2: Audio-transcript system with a described SHAP module; Results report accuracy and a confusion matrix (Fig. 4.1) only. Speech-text setting is eligible, but no explanatory output is shown. [117] X2: SHAP importance is asserted in prose; no coefficient, ranking, plot or instance is reported. [118] X2: EEG/ECG/GSR/speech/face fusion; SHAP and LIME are described, but Figs. 1-4 show architecture and metrics only. No speech explanation output. [119] X2: Descriptor generation is evaluated with caption metrics on EMOSEC; no generated descriptor is shown for an evaluated emotion decision. [120] X3: RAVDESS song subset (sung sentences) only; no spoken-speech analysis. [121] X4: Audio–visual–text fusion; no separate explanation of an audio/audio–transcript SER predictor. KernelSHAP outputs are present; XAI is not absent. [122] X4: SHAP chart includes voice pitch/energy, but explanation is from facial–speech–text HCI fusion, not an independently explained speech/audio–transcript predictor. Text is not clearly speech-transcribed. Figure 5 caption also mislabels its content. [123] X4: LIME word highlights are reported, but only for a fusion model containing facial/video input; no distinct explanation of the audio–transcript ablation was found. [124] X4: Theatre-performance interpretation combines speech with video and gesture; qualifying speech or speech–transcript explanation is not established. Reporting also includes unrelated BRAIN-IP material. [125] X4: SHAP is implemented, but the actual Figures 12–13 report modality-level scores of fused text/audio/visual predictions, not SHAP explanations of a separate speech or audio–transcript SER predictor. Fig.5 is an audio feature map, not an XAI result. [126] X4: The article describes pitch/loudness as SHAP-important (Table 3), but its displayed force plot is explicitly computed using TreeExplainer on a separate XGBoost classifier rather than the claimed transformer fusion SER model. No traceable speech-specific SHAP output from that main model is established. [127] X4: Qualitative attention/modality scores show audio's role in audio–video–text fusion; no independently explained speech/audio–transcript predictor established. [128] X4: EmoSync targets video-based fine-grained multimodal emotion recognition; the audio reasoning pathway is shown in the method but no independently evaluated speech SER explanation is reported. [129] X5: DeepMI explains sample difficulty for curriculum ordering (Sec. V, Fig. 2 onwards), not the emotion decision. [130] X6: KAAP genuinely highlights waveform segments and spoken words, but the displayed core explanations use synthetic TTS derived from written Twitter text and a three-modality predictor; no qualifying explanation on real-speech audio/audio–transcript experiment was demonstrated.